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Bacteria and Bacteriophage Consortia are Associated with Protective Intestinal Metabolites in Patients Receiving Stem Cell Transplantation

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Competing Interests Statement

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Code availability

The scripts and packages used for whole shotgun metagenomic sequencing, viral metagenomic sequencing and multi-omics factor analysis have been deposited to Github (<https://github.com/guardianre/MOFA-in-allo-SCT.git>).

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Abstract

The microbiome is a predictor of clinical outcome in patients receiving allogeneic hematopoietic stem cell transplantation (allo-SCT). Microbiota-derived metabolites can modulate these outcomes. How bacteria, fungi and viruses contribute to the production of intestinal metabolites is still unclear.

We combined amplicon sequencing, viral metagenomics and targeted metabolomics from stool samples of allo-SCT patients (n=78) and uncovered a microbiome signature of *Lachnospiraceae* and *Oscillospiraceae* and their associated bacteriophages, correlating with the production of immuno-modulatory metabolites (IMMs). Moreover, we established the Immuno-modulatory Metabolite Risk Index (IMM-RI), which was associated with improved survival and reduced relapse. A high abundance of short-chain fatty acid biosynthesis pathways, specifically butyric acid via butyryl-CoA:acetate CoA-transferase (*BCoAT*), was detected in IMM-RI low-risk patients and virome genome assembly identified two bacteriophages encoding *BCoAT* as an auxiliary metabolic gene.

In conclusion, our study identifies a microbiome signature associated with protective IMMs and provides a rationale for considering metabolite-producing consortia and metabolite formulations as microbiome-based therapies.

Introduction

Allogeneic hematopoietic stem cell transplantation (allo-SCT) offers a cure for many hematological diseases. Its success is limited by severe morbidity and mortality associated with acute graft-vs-host disease (GvHD).¹ In patients receiving allo-SCT, the intestinal bacterial microbiome is altered by loss of diversity and domination of disease-related taxa.^{2–4} Multiple factors contribute to this GvHD-related microbial signature, such as antibiotics, (radio)chemotherapy and malnutrition.^{5,6} Alterations of bacterial communities influence the pathophysiology of GvHD and affect outcome of patients receiving allo-SCT.^{6–10}

Microbiota-derived metabolites such as short-chain fatty acids (SCFAs), tryptophan-derivatives such as indoles and secondary bile acids (BAs) have been shown to exert immuno-modulatory effects (e.g. via modulation of regulatory T cell [Tregs] function or induction of Type I IFN [IFN-I] signaling) and tissue-homeostatic functions (e.g. by acting on intestinal epithelial cells [IEC]).^{11–13} Propionic and butyric acid, indole-3-carboxaldehyde (ICA) and deoxycholic acid (DCA) have been associated with protection against GvHD in mice and humans¹². Additionally, alterations of bacteria together with metabolites, such as loss of specific anaerobic commensals and a decrease in intestinal SCFA levels, have been linked to GvHD severity in allo-SCT patients.¹⁴

Microbiota-derived metabolites have been analyzed in the blood of acute¹⁵ and chronic GvHD patients¹⁶. However, in humans, the majority of SCFA are metabolized on first pass¹⁷ and while only 2% of intestinal butyric acid becomes systemically available¹⁸, it is essential locally for maintaining intestinal homeostasis where it fuels colonocytes, promotes mucus secretion and enhances expression of tight-junction proteins¹⁹.

The intestinal microbiome is comprised not only of bacteria, but also fungi, viruses, protozoa and archaea. Prokaryotic viruses, i.e. bacteriophages, encoding auxiliary metabolite genes (AMG) can modulate bacterial metabolism while imparting fitness benefits to themselves^{20,21}. Bacteriophage-infected bacteria undergo phage-directed metabolism and exhibit distinct physiology compared to their uninfected peers²². Although analyses of the human gut virome remain challenging due to the high degree of viral-sequence diversity and continual discovery of novel viruses by metagenomics²³, a recent study linked picobirnaviruses to severe GI-GvHD.²⁴ How dynamic shifts in these kingdoms, e.g. after antibiotic exposure (ABX)⁶ or at the onset of GvHD, affects local, intestinal metabolite profiles and contribute to clinical outcome needs to be further investigated.²⁴

Results

Allo-SCT alters bacterial, viral and metabolite profiles

In this prospective, observational, longitudinal cohort we profiled bacterial, fungal and viral communities together with microbiota-derived metabolites in 78 patients receiving allo-SCT at two German academic hospitals (Munich [MUC] and Regensburg [REG]). Starting at Day -7 relative to the day of transplantation (Day 0), weekly stool sampling was performed, resulting in a mean of four samples per patient (Figure 1). Acute leukemia, followed by myelodysplastic syndrome, were the most common indications for transplantation (Supp. Table 1). The majority of patients received reduced-intensity conditioning regimens before transplantation (88% and 87% in MUC and REG, respectively) and were engrafted with unmodified peripheral-blood stem cells (96% and 75%) from unrelated donors (72% and 74%).

Longitudinal profiling of the intestinal microbiome and metabolome demonstrated a loss of diversity in bacterial and viral communities, which was lowest at Day +14 (Figure 2A). Across both centers, fungal diversity did not decline during the peri-engraftment period. In line, bacterial load, quantified by RT-PCR, decreased after allo-SCT while fungal load remained stable (Extended Data Fig. 1A). Viral load was not quantifiable due to the absence of viral house-keeping genes.

Beta diversity analysis revealed that bacterial communities at Day -7 were significantly different from late time-points (Day +7–28, Extended Data Fig. 1B). Likewise, the virome of patients sampled at Day -7 clustered separate from Day +28. For fungome, no differential clustering pattern according to time-point was observed. Identical antifungal therapeutic (but different prophylactic) standards between Munich and Regensburg (Supp. Table 2) did not impact beta diversity (Extended Data Fig. 1C).

Next, we screened for known IMM classes (SCFAs, branched-chain fatty acids [BCFAs], IFN-I inducing metabolites [IIMs] including tryptophan-derivatives and the flavonoid-metabolism byproduct desaminotyrosine (DAT) as well as BAs) in stool samples. By targeted mass-spectrometry (MS), we analyzed a total of 57 intestinal metabolites: Panel 1 consisted of SCFAs, BCFAs, IIMs and lactic acid. Panel 2 consisted of primary (PBA) and secondary BAs (SBA).

At early time-points (Day -7-0), patients had abundant expression of Panel 1 metabolites (Figure 2B & Extended Data Fig. 2A), which declined at late time-points. Lactic acid, which was enriched at late time-points was the exception. In Panel 2, the PBAs cholic acid (CA) and chenodeoxycholic acid (CDCA) were enriched at Day +28, and so were the CA-derived SBAs ursodeoxycholic (UDCA) and tauroursodeoxycholic acid (TUDCA) (Figure 2C & Extended Data Fig. 2B). In contrast, SBAs with known immuno-modulatory functions, such as the CA-derived deoxycholic acid (DCA) as well as the CDCA-derived lithocholic acid (LCA)¹³ were enriched only at early time-points.

Principal component analysis (PCA) showed that metabolite expression profiles at Day -7 were significantly different from late time-points (Extended Data Fig. 2C-D). We observed overlapping clusters between Days +7-28, suggesting that the peri-engraftment state was metabolite-unique and persisted until at least 1 month after allo-SCT.

Bacteria and bacteriophages are associated with immuno-modulatory metabolites

We employed multi-omics factor analysis (MOFA, explained in Supplementary Notes) to integrate four omics-modalities: datasets obtained from (1) bacterial 16S and (2) fungal ITS (internal transcribed spacer 1) amplicon sequencing, (3) viral metagenomic sequencing of isolated virus-like particles (VLP) and (4) intestinal metabolite levels identified and quantified by MS.

By MOFA, we identified complex covariation patterns within and across the bacteriome, virome and metabolome (Figure 3A). In these omics-modalities, approx. 40% of total variance was explained by ten Factors (Figure 3B). In contrast, MOFA explained less than 10% of variance in fungome. Factors 1, 3 and 4 explained a large fraction of variance within the bacteriome, the variance explained within the virome was divided between Factors 1 through 5 and Factors 1 to 3 explained the most variance within the metabolome. In Factor 1, variance observed within the bacteriome and metabolome was explained broadly by a wide range of Features exerting moderate effects (Figure 3C). Since Factors 1 and 3 explained a large proportion of the total variance across the cohort, captured co-variations within bacterial and viral kingdoms as well as metabolites, and were positively correlated with intestinal concentrations of IMMs (Extended Data Fig. 3A), they were the focus for downstream analysis.

Factor 1 was linked to the genera *Ruminococcus*, *Subdoligranulum* and others within the *Oscillospiraceae* family of obligate anaerobes (weight > 1, Figure 3D). Only *Enterococcus* was negatively correlated with Factor 1. All viral Features identified in Factor 1 were bacteriophages, and negatively correlated. Some of these viral clusters e.g., VC334, were predicted to infect *Enterococcus*, and we observed a strong correlation with *Enterococcus*

(Extended Data Fig. 4A). Panel 1 metabolites had the highest Feature weights and were positively correlated, including the SCFAs butyric, propionic and valeric acid, the BCFAs 2-methylbutyric, isovaleric and isobutyric acid as well as the tryptophan-metabolite ICA. Panel 2 metabolites had both positive – with the immuno-modulatory SBAs DCA and the LCA derivative Dehydro-LCA –, and negative correlations – with the SBAs UDCA and TUDCA.

Factor 3 was linked to the genera *Lachnoclostridium*, *Anaerostipes*, *Blautia* and others from *Lachnospiraceae*, a family of obligate anaerobes that ferment plant polysaccharides (weight > 0.5, Figure 3E). As in Factor 1, all top viral Features were bacteriophages. In contrast to Factor 1, the viral clusters BAC3, VC62, VC64 and VC27 were positively correlated, and associated with genera within the *Lachnospiraceae* family (Extended Data Fig. 4B). Factor 3 was positively correlated with many Panel 2 metabolites, including the SBA ursocholic acid, the oxidized SBAs 3- and 7-dehydrocholic acid (DHCA) as well as the PBAs CA and CDCA. From Panel 1, we detected a positive correlation with DAT, SCFAs and BCFAs.

Even though Factor 4 only explained a small fraction of variance in the metabolome, we noted the covariation of *Bacteroides* and *Parabacteroides* with the viral clusters VC191 and VC612 together with propionic acid levels (Extended Data Fig. 3B & Extended Data Fig. 4C).

MEFISTO²⁵ is a toolbox for factor analysis of multimodal data with spatial or temporal dependencies. Given that we analyzed a longitudinal cohort, temporal dependencies existed between samples. Therefore, we employed MEFISTO to assess the extent to which the variation captured by the Factors could be attributed to time. The highest scale value assigned by MEFISTO was 0.19 for Factor 1, followed by 0.05 for Factor 7, suggesting only a low time-dependency of the Factors (Extended Data Fig. 3C).

For both Factors, the top 15 viral Features identified by MOFA were bacteriophages, which target bacterial hosts. Indeed, when we queried the intestinal virome across our cohort, we confirmed that bacteriophages were in much higher abundance than eukaryotic viruses – including human DNA viruses (Supplementary Figure 1) –, both before and after allo-SCT (Extended Data Fig. 3D). In addition, our viral metagenomic sequencing pipeline identified numerous unclassified viral contigs.

In light of recent reports suggesting that bacteriophages can modulate metabolic processes in bacteria²¹, we correlated the top bacterial and viral Features and linked their abundance to the normalized concentration of intestinal IMMs. We observed significant correlations with a Pearson's correlation coefficient ranging from R=0.5 to R=0.7 for genera within the *Oscillospiraceae* and *Lachnospiraceae* families including *Ruminococcus* and *Blautia* with the bacteriophages BAC3, VC62 and VC27. Intestinal metabolite concentrations, shown exemplary for butyric acid (Figure 3F) were high when bacteria and bacteriophages were co-abundant. Likewise, we observed high intestinal concentrations of propionic acid, the BCFA isovaleric acid as well as the IIMs DAT and ICA when both bacteria and bacteriophage were present in high abundance (Extended Data Fig. 5A–D).

In sum, MOFA-identified Factors 1 and 3 described a functional microbiome signature linking intestinal SCFA, BCFA and IIM with bacterial taxa from the *Oscillospiraceae* and *Lachnospiraceae* families and bacteriophages.

The immuno-modulatory metabolite risk index (IMM-RI) predicts outcome

The intestinal microbiome is a predictor of outcome in allo-SCT patients⁷. Thus, we asked whether the microbiome signature defined by MOFA Factors 1 and 3 associated with outcome. We observed that low Factor 1 and 3 or either together with high Factor 4 identified individual patients that were deceased due to TRM at a 2-year follow-up (Figure 4A–C). Once we determined optimal Factor value cut-points via the Youden index, we noted that increased Factors 1 and 3 or decreased Factor 4 were significantly correlated with OS by log-rank test (Supp. Table 3).

The mechanism that links bacterial diversity with outcome is not well understood. We hypothesized that this may be in part due to a depletion of beneficial metabolites and therefore we assessed intestinal metabolite levels sequentially starting at Day –7 before allo-SCT (baseline). In addition to MOFA-identified metabolites, we included BAs that have been previously described to attenuate GvHD (UDCA²⁶, TUDCA²⁷) or regulate IFN-I responses (DCA, LCA)²⁸ in our analysis. IMM levels declined progressively after allo-SCT: At Day +21 their median intestinal concentration was substantially reduced compared to baseline, e.g.: butyric acid (162-fold), propionic acid (68-fold), isovaleric acid (149-fold), DAT (53-fold) and ICA (475-fold) (Figure 4D). Likewise for the IIMs DCA (73-fold) and LCA (41-fold). Analogous to metabolite expression profiles (Extended Data Fig. 2C–D), depletion of these IMM levels persisted until at least 1 month after allo-SCT. In contrast, at Day +28 we observed a significant increase in the PBAs CA (40-fold) and CDCA (430-fold) as well as the SBAs TUDCA (313-fold) and UDCA (26-fold) compared to baseline (Extended Data Fig. 5E).

We hypothesized that low levels of IMM levels might be deleterious in allo-SCT patients. Considering (1) the shared microbial biosynthesis pathways for production of different SCFA molecules (e.g. regulation of acetate and propionate metabolism via the TCA cycle²⁹) (2) the detection of taxa associated with *both* SCFA and BCFA (Extended Data Fig. 4), but (3) the distinct pathways involved in SCFA, tryptophan-derivative and DAT synthesis, we developed an “Immuno-modulatory Metabolite Risk Index” (IMM-RI) to associate metabolites with clinical outcomes. We established the IMM-RI by compounding intestinal levels of the SCFAs butyric and propionic acid, the BCFA isovaleric acid, and the IIMs DAT and ICA. These MOFA-identified metabolites were selected based on significant associations with OS (Supp. Table 4), or in the case for butyric acid, prior studies reporting a negative correlation with adverse outcomes in allo-SCT¹⁴.

First, we confirmed that patients with above-median bacterial diversity had significantly higher overall survival (OS), less transplant-related mortality (TRM) and reduced incidence of GI-GvHD (Figure 5A, Extended Data Fig. 6A–C) at a follow-up of two years. This established that observations in our cohort were comparable to other international cohorts⁷. In contrast, neither fungal nor viral diversity correlated with clinical outcome.

In the Kaplan-Meier analysis, the IMM-RI effectively characterized 2-year OS, with 89.5% (standard error (SE) 7%) survival in the low-risk compared to 47.6% (SE 9%) in the high-risk group (HR 6.2, 95% CI 1.4–26.9, Figure 5B, Supp. Table 3). To our surprise, the IMM-RI was significantly associated with relapse: the cumulative incidence of relapse was elevated in the IMM-RI high-risk group taking into account TRM as competing risk (Figure 5C). TRM was also independently correlated with the IMM-RI. Cumulative incidence of GI-GvHD was elevated in the IMM-RI high-risk group with death as competing risk (Figure 5D). With respect to mortality, the IMM-RI demonstrated an AUC of 0.766 (95% CI 0.634–0.897, Figure 5E) with a negative predictive value of 0.895 and positive predictive value of 0.516. All corresponding hazard ratios are reported in Supp. Table 3.

Patients with low-risk IMM-RI had significantly higher Factor 1 and 3 values (Figure 5F), suggesting an association between these two functional microbiome read-outs.

SCFA synthesis pathways are abundant in IMM-RI low-risk patients

To divulge the microbial pathways which enabled IMM-RI low-risk patients to sustain IMM production during the peri-engraftment period, we performed whole shotgun metagenomic sequencing in a subset of IMM low- and high-risk patients. The most significant and differentially abundant microbial pathways involved the degradation of 4-aminobutanoate to butyric acid (PWY-5022), the biosynthesis of thiamine diphosphate (vitamin B1, PWY-7282), starch degradation III (PWY-6731), the degradation of N-acetylneuraminate to acetic acid or ethanol (P441-PWY) as well as histidine degradation (PWY-5030 & HISDEG-PWY) (Figure 6A–B). These and related pathways were detected predominantly in *Flavonifractor plautii* of the *Oscillospiraceae* family, matching MOFA-identified bacteria (Extended Data Fig. 7A).

In IMM-RI low-risk patients, we identified an increased abundance of metabolic pathways involved in starch and amino acid (AA, including proteinogenic and non-proteinogenic) degradation (Figure 6C, Extended Data Fig. 7B). Via the pathway PWY-5022, bacteria can degrade the non-proteinogenic AA 4-aminobutanoate (GABA), an inhibitory neurotransmitter, to butyric acid. Since we noted that various pathways related to fermentation were highly differentially abundant (Figure 6B), we combined these into a superclass of fermentation which was significantly increased in IMM-RI low-risk patients (Figure 6D). Since SCFA are derived from the fermentation of dietary fibers, complex carbohydrates including starches or AA, we specifically analyzed a superclass of SCFA synthesis, which includes biosynthesis pathways of acetic, butyric and propionic acid (Figure 6E, Supp. Table 5). We observed an increased abundance of the SCFA superclass in IMM low-risk patients, which corresponded to a significant increase in abundance in butyric acid and nonsignificant elevation in acetic and propionic acid biosynthesis pathways (Figure 6E, Extended Data Fig. 7C).

Besides SCFA biosynthesis, we detected a higher abundance of pathways involved in AA synthesis (including essential and non-essential AA), AA degradation (histidine), purine nucleotide degradation (including adenosine and guanosine) and B vitamin synthesis (Extended Data Fig. 7B) in IMM-RI low-risk patients. Degradation of L-histidine via the HISDEG-PWY can lead to production of the metabolite imidazole propionate, which

was shown to modulate host inflammation and metabolism³⁰. Purine metabolism was recently linked to efficacy of immunotherapy³¹, while B vitamin synthesis³² was associated with protection from colitis. Genes involved in the conversion of PBA to SBA were not differentially abundant.

Butyric acid is synthesized via the intermediates acetyl coenzyme A (CoA) and butyryl CoA. Synthesis is completed via the enzymes butyrate kinase or butyryl-CoA:acetate CoA-transferase (encoded by the gene *BCoAT*)³³. Both genes were more abundant in IMM-RI low-risk patients (Figure 6F–G). We quantified *BCoAT* copy numbers via quantitative polymerase chain reaction analysis (qPCR). Patients in which *BCoAT* was detected had a significantly higher Factor 1 value (Figure 6H). The detection of *BCoAT* was reduced in the peri-engraftment period (Figure 6I) and copy numbers declined significantly in paired patient samples after Day –7 (Figure 6J). We observed significantly reduced *BCoAT* copy numbers in patients exposed to antibiotics (ABX, Figure 6K), confirming our previous findings that antibiotics are an independent risk factor for low *BCoAT* copy numbers³⁴.

Bacteriophages encode the butyric acid synthesis gene *BCoAT*

Bacteriophages can modulate bacterial metabolism via AMGs²¹, which are activated when phages infect their hosts, and then augment or redirect specific metabolic pathways within them^{20,21}. To identify phage-encoded AMGs, we mined our viral metagenomic sequencing dataset for the microbial pathways and genes identified above. We detected whole length (1.34 kb) and a 450 bp fragment of *BCoAT* encoded in two individual bacteriophage contigs with almost complete phage genome length: VC-2 (90,756 bp) and VC-1 (77,772 bp). Phylogenetic analysis showed a close evolutionary relationship of the bacteriophage-coded *BCoAT* to their homologies from *Oscillospiraceae*, indicating that this gene may have been acquired from this bacterial family (Figure 7A). In addition, we discovered *BCoAT* in a *Myoviridae* sp. bacteriophage (GenBank ID: DAV55430.1) identified in another cohort of human metagenomic samples including those from the Human Microbiome Project³⁵, indicating that the two *BCoAT*-coding phages identified in this study are unlikely to be an assembly artefact.

We aligned *BCoAT* coded in VC-2 against a closely-related bacterial reference sequence from *Lawsonibacter* (a species within *Oscillospiraceae*) and observed a close match in sequence identity for both *BCoAT* and the adjacent DEAD box helicase (Figure 7B, Extended Data Fig. 8A), which in bacteria are involved in ribosome biogenesis and translation initiation³⁶. Meanwhile, VC-1, which encoded a part of the *BCoAT* gene, aligned with *Pseudoflavonifractor*, another member of *Oscillospiraceae* (Extended Data Fig. 8B).

Genomic assembly and annotation characterized VC-2 and VC-1 as temperate bacteriophages, evident by the presence of genes for integration, excision and packaging (Figure 7C). Phage AMGs are frequently located within the auxiliary gene cassettes and are separated by structural and nucleotide metabolism cassettes^{20,37}. The organization of *BCoAT* in the genomes of VC-2 and VC-1 was consistent with this: the tail (structural) gene cassettes were clustered together, but located separately from the *BCoAT* AMG.

Patients which harbored VC-2 and VC-1 had significantly elevated Factor 1 values (Extended Data Fig. 8C–D). We observed that VC-2 and VC-1 were more abundant and more likely to be detected in IMM-RI low- vs high-risk patients (Figure 7D–E, Extended Data Fig. 8E–F). Despite their low relative abundance, we detected VC-2 and VC-1 in more than 60% and 80% of patients across both centers at Day –7, respectively, but detection declined progressively during the peri-engraftment period. Exposure to antibiotics resulted in a significant decrease of the relative abundance of VC-2 and VC-1. The abundance of both bacteriophages was moderately correlated with intestinal butyric acid levels by Spearman's correlation (Figure 7F).

Onset of GI-GvHD and exposure to antibiotics impact IIMs

Clinical events during the peri-engraftment period, such as onset of GI-GvHD and initiation of broad-spectrum ABX, have been shown to compound the detrimental effect of allo-SCT on the bacterial microbiome^{6,7,38} and may impact intestinal IIMs.

We performed a subgroup analysis in patients with clinical diagnosis of GI-GvHD of any severity matched by time-point to a control group of allo-SCT patients without GvHD. In GI-GvHD patients, we observed a trend towards lower bacterial alpha diversity and load as well as a significant decline in viral richness compared to controls (Figure 8A, Extended Data Fig. 9A). We did not observe any differences in fungal diversity or load (Extended Data Fig. 9A–B). Beta diversity analysis showed significantly distinct clustering of bacterial and viral, but not fungal communities, in GI-GvHD patients compared to controls (Extended Data Fig. 9C).

GI-GvHD patients presented with significantly lower levels of IMM-RI metabolites (Figure 8B). Only ICA showed a non-significant reduction. Other IIMs such as acetic, valeric, 2-methylbutyric and isobutyric acid were also significantly reduced (Extended Data Fig. 9D). Of BAs, only LCA was significantly decreased in GI-GvHD patients compared to controls, while CA, CDCA, UDCA and TUDCA levels were comparable between both groups (Extended Data Fig. 9D).

During the course of their hospital stay, 100% of MUC and 94% of REG patients were started on broad-spectrum ABX (Supp. Table 6) as per center-specific standards (Supp. Table 2). We longitudinally analyzed matched patient samples before and after ABX-exposure. ABX resulted in a significant decrease of bacterial alpha diversity and load as well as viral richness (Figure 8C & Extended Data Fig. 10A). Fungal diversity or load was not impacted by ABX (Extended Data Fig. 10A–B). By beta diversity analysis, we observed that ABX resulted in major and significant changes to bacterial and viral, but not fungal, community compositions (Extended Data Fig. 10C).

ABX drastically affected the metabolites comprising the IMM-RI and resulted in their depletion to levels approaching the lower limit of detection (Figure 8D). Other IIMs were also significantly diminished (Extended Data Fig. 10D). Likewise, MOFA Factor 1 and 3 values declined significantly after ABX-exposure (Figure 8E). In contrast, we observed that the BAs CA, UDCA and TUDCA were significantly increased in patients after ABX exposure (Extended Data Fig. 10D). At species level, we observed that ABX-exposed

patients had a higher abundance of *Enterococcus* (Figure 8F) but a decrease in species predominantly within the *Lachnospiraceae* and *Oscillospiraceae* families, many of which corresponded to high-weight bacterial Features identified by MOFA e.g., *Blautia wexlerae* or *Flavonifractor plautii*.

Discussion

We identified a functional microbiome signature comprised of bacteria within the *Oscillospiraceae* and *Lachnospiraceae* families and their associated bacteriophages. This signature was linked to the production of intestinal SCFAs, BCFAs and IIMs, which were associated with increased OS, reduced TRM and less relapse after allo-SCT. Recently, defective oxygen utilization in IECs was linked to the loss of intestinal hypoxia during GvHD³⁹. This supports our finding that the obligate anaerobes *Oscillospiraceae* and *Lachnospiraceae* were associated with IIMs, which together may exert local, protective effects on IECs; and that these consortia and their metabolites are depleted during GvHD when a shift to an aerotolerant microbiome occurs.

MOFA-identified bacteria included *Roseburia*, which are known butyrate producers and affect colonic motility, maintain immunity and exert anti-inflammatory properties⁴⁰. *Flavonifractor plautii*, a species linked to Tregs induction in mice after oral administration⁴¹, comprised Factor 3 and was associated with butyric acid production via *BCoAT*. We characterized bacteriophages that were tightly correlated with their *Oscillospiraceae* or *Lachnospiraceae* hosts, and where both were highly co-abundant, we observed high-level expression of intestinal IIMs. This association between bacteriome, virome and metabolites may suggest an important role for bacteriophages in synthetic consortia engineered to produce metabolites. However, it remains to be determined through *in vitro* follow-up studies whether bacteriophages are essential for metabolite production or if they simply coexist with their respective hosts.

Our viral metagenomic sequencing pipeline, which was optimized for non-targeted DNA sequencing from isolated VLPs and utilizes large sample masses to avoid bias associated with whole genome amplification, allowed us to observe a high abundance of bacteriophages in patients, which declined along with intestinal bacterial diversity after allo-SCT. Our results partially align with a study of the intestinal virome in 44 patients after allo-SCT which found an increase of eukaryotic viruses and a decrease of bacteriophage richness²⁴. Although there were no major alterations in the intestinal eukaryotic virome and we detected human DNA viruses such as *Poxiviridae* and *Papillomaviridae*, we acknowledge that our viral sequencing pipeline was not enriched for RNA-virus detection. However, given that bacteria make up more than 50% of total solids in human feces⁴², our findings suggest that virome alterations during allo-SCT occur at the level of bacteriophages, which shift in response to factors affecting their host bacteria. This is consistent with a recent MOFA in patients with sepsis at the intensive care unit, which showed that bacteria and their bacteriophages co-varied in Factors driven by antibiotic perturbation⁴³.

Fungal diversity and load by ITS sequencing remained stable during allo-SCT and did not decline in GI-GvHD patients⁴⁴. The low MOFA-explained variance in fungome suggested

that MOFA considered most of the variations within this dataset to be random and not driven by specific feature covariation patterns that might represent distinct biological states. In contrast, culture-based studies have identified a subset of patients in which *Candida parapsilosis* complex species were correlated with worse OS and higher TRM⁴⁴, and found that colonization by *Candida spp.* was associated with more severe GvHD⁴⁵. For associations with clinical outcome, traditional culture-based methods remain the standard.

Microbiota-derived metabolites have been shown to protect from GvHD in animal models.^{11,12,46,47} Previous studies in allo-SCT patients have been cross-sectional and focused on metabolites in the blood.^{15,48} An untargeted MS-based screening of metabolites in the serum of allo-SCT recipients and donors in French transplant centers found that at the onset of GvHD, tryptophan-derivatives and other aryl hydrocarbon receptor (AhR) ligands were decreased.¹⁵ In line, we observed a trend towards reduction of the tryptophan-derivative ICA in GI-GvHD patients and a significant reduction in allo-SCT patients during the peri-engraftment period. In the same study, the serum SBAs UDCA and TUDCA were increased in patients with acute GvHD. We observed a significant increase of intestinal UDCA and TUDCA at late time-points after allo-SCT and in patients exposed to ABX, but noted no differences in GI-GvHD patients compared to controls. The use of peroral UDCA to prevent hepatic complications and severe acute GvHD in allo-SCT patients²⁶ may explain elevated levels of fecal UDCA at late time-points (Supp. Table 2).

Besides SCFAs, we observed that allo-SCT impacted expression of several intestinal metabolites which were not previously described in the human allo-SCT setting, such as DAT⁴⁹, ICA¹², LCA⁵⁰ and DCA⁵¹. These metabolites – which we refer to as IIMs – are potent activators of IFN-I signaling, which can positively or negatively modulate GvHD depending on the timing of IFN-I activation⁵². DAT, also known as phloretic acid, a byproduct of flavin degradation, can enhance IFN-I responses and antiviral activity in murine respiratory influenza infection.⁴⁹ We detected ICA, which has been shown to target AhR and protect mice from metabolic syndrome⁵³ as well as GvHD¹² via IFN-I.¹² We measured DCA, which has been shown to restrict alphavirus infection in mice via IFN-I.⁵¹ The SBA LCA was decreased in GI-GvHD patients, and two recent studies have associated SBAs with Tregs differentiation^{47,54}, which play a vital role in restraining GI-GvHD. Recently, intestinal levels of DAT together with DCA, LCA and other SBAs were compounded in a “microbiome metabolic profile” and associated with outcome in critically-ill COVID-19 patients⁵⁵. In our study, intestinal levels of IMM-RI metabolites declined progressively after allo-SCT, and these changes persisted until at least 1 month after allo-SCT. In GI-GvHD patients, IMM levels were particularly low, but further conclusions are limited by the number of patients in these two subgroups (n=26 and n=16 for GI-GvHD and controls, respectively).

The IMM-RI predicted OS and was independently associated with reduction of TRM and relapse rate. While our non-interventional study is limited to associating microbiome signatures and metabolites with clinical outcome after allo-SCT, there are plausible mechanisms by which the metabolites that constitute the IMM-RI might impact outcomes. A decrease in TRM may reflect the observed reduction in the incidence and severity of GvHD, and aligns with the reduction of butyric acid levels reported in severe GI-GvHD patients¹⁴.

Indeed, Tregs, which can be activated by metabolites including SCFAs⁵⁶, play a crucial role in controlling GvHD¹¹. Glucagon-like peptide 2 (GLP-2) was shown to reduce GI-GvHD by promoting IEC regeneration but had no effect on graft-vs-leukemia (GvL)⁵⁷. Therefore, the reduced relapse rate in IMM-RI low-risk patients may be dependent on other metabolites besides SCFAs, such as IIMs, which could augment IFN-I signaling and invoke GvL in patients⁵⁸. The effects on both TRM and relapse may be interpreted as a cumulative effect of intestinal metabolites with different underlying modes of action.

Only few studies have reported an association between the intestinal microbiota and relapse after allo-SCT.⁵⁹ As expected, bacterial alpha diversity correlated with OS but did not predict relapse rate. However, a low IMM-RI was significantly associated with reduced relapse rate considering its competing risk TRM. We detected no significant differences in regard to sex, HCT-CI, disease risk, conditioning intensity, exposure to antibiotics and other parameters between IMM-RI low- and high-risk patients (Supp. Table 7). Old age correlated with high-risk IMM-RI, which is consistent with age being a risk factor for an increase of biomarkers of acute GI-GvHD³⁸ and with age-related changes of intestinal microbiota promoting intestinal inflammation and loss of epithelial integrity⁶⁰. However, this association between intestinal IIMs and relapse rate was unexpected, and our study was not *a priori* designed for this purpose. The number of patients to calculate the IMM-RI was limited (n=50) and few relapse events in this subset (only 9 vs 1 events in the high-risk vs low-risk IMM-RI groups, respectively) precluded a multivariable analysis. Therefore, these results should be considered exploratory findings and require confirmation in an international cohort sufficiently powered for multivariate analysis.

Microbial pathways for the biosynthesis of SCFAs were more abundant in IMM-RI low-risk patients, and these results were confirmed for butyric acid biosynthesis by qPCR for *BCoAT*, a gene encoding a key enzyme in the terminal reaction of the acetyl-CoA pathway – the predominant biosynthesis pathway of butyric acid⁶¹. Strikingly, we identified *BCoAT* encoded in two temperate bacteriophages, VC-2 and VC-1. Environmental stressors, such as ABX (e.g., mitomycin C⁶²), can trigger temperate bacteriophages to change their behavior and initiate a lytic cycle. This cycle results in lysis of host bacteria, release of new phage particles and an increased abundance of host-originated genes, including AMGs, with each subsequent infection. On the other hand, during lysogeny, temperate bacteriophages are stably integrated into the host genome, allowing them to coexist and multiply alongside host cells. Since bacteriophages do not require metabolism genes, it stands to reason that detection of the *BCoAT* AMG in numerous patients across both centers may be due to a selective pressure which confers an advantage to colonize certain niches. Our data provides intriguing evidence of bacteriophage AMGs in humans, highlighting the potential cooperation of bacteria and their bacteriophages via which they could support metabolite production and potentially benefit patients receiving allo-SCT.

Besides microbiota, factors relating to host genetics, diet and medication contribute to intestinal metabolite profiles. All MUC patients received in-patient germ-reduced Western pattern fare with a defined macronutrient composition and goal intake of 2,000 calories/day (Supp. Table 8). However, associations between diet and metabolites are limited since we did not record patient-individual diets.

Antibiotics impact intestinal bacteria.^{2,34} Loss of anaerobic commensals and decrease of intestinal SCFA levels, specifically acetate, propionate and butyrate, were associated with severe acute GvHD¹⁴. Intestinal butyrate and propionate were negatively correlated with exposure to antibiotics⁶³. By longitudinally tracking patients from before allo-SCT, we were able to analyze in paired patient samples how initial exposure to antibiotics depleted protective IMMs. However, despite ABX's dominant effect on microbiome, other factors such as concomitant medication or diet may have changed between sequential samples and could contribute.

Our findings have far-ranging implications that could be important for other T cell mediated therapies that are influenced by the gut microbiome, such as CAR-T cells or immune checkpoint inhibitors (ICI)^{64,65}. Firstly, administration of IMMs – either by synthetic bacterial/bacteriophage consortia engineered to produce specific metabolites, or directly via metabolite formulations – could be explored in clinical trials in allo-SCT patients with biomarker-proven intestinal microbiota disruption (e.g., IMM-RI high-risk patients) to treat or prevent GvHD and relapse. Secondly, screening for IMMs in FMT products could assist in donor-recipient matching, standardization of product quality and development of a precision FMT product that may improve response rates. Thirdly, we have previously shown that timing of ABX impacts the intestinal bacteriome⁶, and it stands to reason that limiting ABX exposure or altering its timing could be used to modulate intestinal IMM levels and with it, clinical outcome.

In clinical scenarios, a functional microbiome snapshot via rapid metabolite measurement would be advantageous over a protracted metagenomics approach. Thus, developing in-hospital assays for a specific index with a limited number of metabolites, such as the IMM-RI, appears feasible.

In sum, our research provides a strong rationale for the design of precision bacterial and viral consortia with desired metabolite profiles (IMMs) as a new frontier in microbiome-based therapies aimed at improving clinical outcomes after allo-SCT and potentially other T cell mediated therapies.

Methods

Study Design

This research complies with all relevant ethical regulations. Study protocols were approved by the institutional review boards (IRB) “Ethikkommission der TU München” (295/18 S) and “Ethikkommission der Universität Regensburg” (14–47_1–101, 14–101-0047, 21–2521-01).

We report the 2-year follow-up of a prospective, observational, longitudinal, two-centric cohort of patients receiving allo-SCT. We combined microbiome analysis with targeted metabolomics to characterize intestinal bacterial, fungal and viral community composition together with the intestinal metabolome. The bacteriome and fungome were characterized via 16S and ITS1 amplicon sequencing, respectively. For virome, we first purified virus-like

particles (VLP), followed by whole genome shotgun sequencing.⁶⁶ We applied MS for quantification of microbiota-derived metabolites.

Patients were enrolled at the University Hospital of the Technical University of Munich (MUC) and the University Hospital Regensburg (REG) after obtaining informed consent and in accordance with IRB-approved study protocols. Patients did not receive financial participant compensation.

Patient characteristics including numbers, age, sex, diagnosis and survival are exhibited in Supp. Table 1. The study cohort included female and male patients. Sex (biological attribute) was assigned based on electronic patient records. We did not consider sex in the study design and did not perform separate analysis according to sex, since sex did not affect microbiome-predicted outcomes in other cohorts of allogeneic stem-cell transplantation patients.^{7,67} In Regensburg, ethnicity was determined by self-reporting, all patients identified as “Non-Hispanic/white”.

Samples were collected once a week beginning at Day -7 before allo-SCT up until Day +35 after allo-SCT or when patients were discharged, whichever occurred later. In the event of occurrence of acute GI-GvHD, patients were sampled and weekly thereafter until resolution of GI-GvHD or death. Samples were obtained on one to two days around the specified time-points. Stool samples were collected in magiX PB1 microbiome preservation buffer (microBIOMix GmbH, Regensburg) or as unprocessed, native samples and transferred to the laboratory within four hours where they were stored at -80 °C until further processing.

Graft-vs-host Disease

GI-GvHD occurred in 50% of patients, of which 30% developed severe GI-GvHD, defined as GI-GvHD grade 2 to 4. At occurrence of GvHD, patients received i.v. corticosteroid therapy. Stool samples of patients with active GI-GvHD were labeled as such, and compared to time-point matched samples from patients without GvHD (control allo-SCT patients). When more than one sample per patient was available for GI-GvHD vs. no GI-GvHD comparisons, the mean metabolite concentration was calculated.

Antibiotics

Patient samples were grouped according to antibiotic status: Samples categorized as “no ABX” had no exposure to antibiotics, but once a patient was started on antibiotics the first and all subsequent samples were classified as “ABX”. In addition, all samples were annotated according to concomitant anti-infective prophylaxis (site-specific standards, see Supp. Table 2), as well as antifungal and antiviral therapy (Supp. Table 6). When more than one sample per patient was available for ABX vs. no-ABX comparisons, the mean metabolite concentration was calculated.

Immuno-modulatory Metabolite Risk Index

The immuno-modulatory metabolite risk index (IMM-RI) was developed in analogy to the “microbiome metabolite profile” in COVID-19 patients⁵⁵. Five metabolites were selected according to univariate association with outcome (propionic acid, isovaleric acid, DAT and

ICA) or biological plausibility (butyric acid). Metabolite levels at Days +7–21 were used for the analysis. When more than one sample per patient was available, the mean metabolite concentration was used. Optimized thresholds for concentrations of individual metabolites were determined using the Youden Index (Supp. Table 9). If concentrations were beneath the cut-off, one point was assigned to that metabolite. A minimum of 0 points and a maximum of 5 points could be assigned to the IMM-RI. Patients were stratified into high and low IMM-RI groups according to their score (0–2: low IMM-RI, 3–5: high IMM-RI).

Analysis of Clinical Outcomes

MOFA Factors: Patients were stratified according to peri-engraftment (Days +7–21) Factor values for Factors 1, 3 and 4. Optimized thresholds were determined using the Youden Index. The effect of Factor values on OS was assessed via log-rank test, and hazard ratios were reported using the Cox regression model (Supp. Table 3).

Alpha diversity: Patients were stratified into higher- and lower-diversity groups (above and below median diversity, respectively) according to the center-specific median inverse Simpson's diversity index during the peri-engraftment period. This practice is consistent with previous studies⁷. When more than one sample per patient was available, the mean inverse Simpson's diversity index was used. OS was analyzed using the Kaplan–Meier estimator. Cumulative incidences of relapse and transplantation-related mortality (TRM), as well as GvHD and death were estimated accounting for the competing risks of each of the respective endpoints (relapse vs. TRM and GvHD vs. death) in R based on the *cmprsk* package.⁶⁸

IMM-RI: Patients were stratified into high and low IMM-RI groups. Cumulative incidence functions between groups were tested for equality using the Gray's test⁶⁹ and the hazard ratio for the different endpoints was calculated in R based on the Fine and Gray model using the *crr* function⁷⁰ (TRM, GvHD) and the Cox regression model (OS) using SPSS 29.

Statistics and Reproducibility

All patients receiving allo-SCT in MUC (between 2019–2021) and REG (between 2018–2021) were screened, and patients with a minimum of three longitudinal samples were included in the analysis. Samples obtained after Day +100 after allo-SCT (n=2) were excluded from the analysis, since the microbiome at these late time-points is not comparable to that of the early post-transplantation period. No statistical method was used to pre-determine sample sizes but the size of our cohort (n=78) is similar to those reported in previous publications.⁶⁷ Randomization did not apply in our non-interventional cohort. Data collection was not performed blind and investigators were not blinded to outcome assessment. Alpha diversity metrics and metabolite levels were neither normally nor lognormally distributed as assessed by Shapiro-Wilk testing. Microsoft Excel and IBM SPSS 29 were used for calculations and analyses. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Bacteriome and Fungome Analysis

Extraction of nucleic acids: 50 mg stool were mixed with a pool of three spike bacteria *Alcanivorax borkumensis*, *Agrobacterium radiobacter*, and *Alicyclobacillus acidiphilus* to 2×10^7 , 1.6×10^7 and 6×10^8 16S rDNA copies per sample, respectively, which served as internal process controls.⁷¹ Cells were lysed by applying a repeated bead-beating protocol using 0.5 mm yttria-stabilized zirconium oxide beads (MP Biomedicals, Eschwege, Germany) on a TissueLyser II instrument (Qiagen) followed by Proteinase K digestion (Roche, Mannheim, Germany). DNA was purified by means of the MagNA Pure 96 instrument (Roche) using the MagNA Pure 96 DNA and Viral NA Large Volume Kit (Roche).

Amplification and sequencing of V1–3 16S rDNA and fungal ITS1 regions:

V1 to V3 hypervariable regions of bacterial 16S rRNA genes were amplified from a total of 1×10^7 bacterial 16S rDNA copies for each sample using the forward primer S-D-Bact-0008-c-S-20 containing sample-specific barcode and sequencing adaptor sequences together with reverse primer S-D-Bact-0517-a-A-18 containing a 3'-P1 adapter sequence. Fungal Internal transcribed spacer regions 1 (ITS1) were amplified from 5 ng DNA using primer pairs ITS1–30F/ITS1–217R⁷² containing barcode and sequencing adaptors. The resulting amplicons were purified with MagSi-NGSPREP Plus beads (Steinbrenner Laborsysteme, Wiesenbach, Germany). Amplicon copy numbers were determined using the KAPA Library Quantification IonTorrent Kit (Roche). DNA library was prepared by pooling equimolar concentrations of adaptor-labeled amplicons. Re-amplification of the final sequencing library by isothermal amplification, chip loading and high-throughput sequencing was carried out using the IonChef™ instrument and the IonTorrent™ Genestudio S5 Plus sequencer (Thermo Fisher Scientific, Waltham, USA).

Quantification: Bacterial and fungal DNA was quantified by real-time PCR using pan-bacterial (16S) and pan-fungal (28S) specific primers, respectively, and amplicon copy numbers were normalized to gram of stool. Quantification of 16S rDNA copies by qPCR was carried out as published.⁷³ Fungal load was quantified by using primers NL1 and 260R and a 28S rDNA-specific FAM-labelled hydrolysis probe.⁷⁴ DNA fragments of *Penicillium roqueforti* 28S rDNA amplified with NL-1/NL-4 primers which were cloned into the pJET1.2 vector (Thermo Fisher Scientific) were used as quantification standard.

Data Analysis: Raw sequencing reads were subjected to Trimmomatic⁷⁵ (0.39) for sliding window trimming applying a windows size of 25 and a quality cutoff of 20. Cutadapt⁷⁶ (3.4) was used for demultiplexing and removal of sequencing adapters and 16S-specific PCR primer sequences. Zero-radius operational taxonomic units (zOTUs) were generated in R (4.2) from demultiplexed reads using a vsearch-based workflow⁷⁷. Standard parameters were used, except an expected error cutoff of 5 and an alpha value of 2. Taxonomy of zOTUs was predicted by the IDTAXA algorithm of the DECIPHER⁷⁸ package after matching to the All-Species Living Tree 16S reference database release 01_2022 (bacteria). The February 2020 release of the Unite database⁷⁹ (fungal ITS1) was used to predict fungal taxonomy. Alpha- and beta-diversity analyses were conducted applying the phyloseq⁸⁰ package version 1.36. Species Richness was represented by summarized zOTUs. Means of groups were compared with the Wilcoxon Rank Sum test (two-sided) adjusted for multiple

comparisons via the Benjamini & Hochberg procedure. Beta diversity was calculated by weighted Unifrac and distances between time-points or groups (No ABX vs ABX; No GI-GvHD vs GI-GvHD) were projected in a principal coordinate analysis (PCoA). Principal coordinates were calculated from weighted UniFrac distances after cumulative sum scaling of zOTU counts. Significance of group differences were analyzed by PERMANOVA using the pairwiseAdonis⁸¹ (0.4) package. Differential abundant features were identified with Maaslin2⁸² 1.7.3 after total sum scaling and log-transformation of zOTU counts. An alpha-level of 0.05 was applied to p-values which were adjusted by the Benjamini & Hochberg method. Individual patients were applied as random effects for calculating the linear model. Plots were generated by ggpubr (0.4).

Whole Shotgun Metagenomic Sequencing: DNA was extracted from stabilized stool samples as described above with the exception of adding bacterial spike controls to the sample. A total of 150 ng of DNA was used for sheared to mean 150 bp using the M220 focused-ultrasonicator (Covaris) followed by size selection with the MagSi-NGSPREP Plus beads (Steinbrenner Laborsysteme, Wiesenbach, Germany) using 1.1x and 1.7x DNA to beads ratio. The Ion Plus Fragment Library Kit (Thermo Fisher Scientific, Waltham, USA) was used for end repair and ligation of sequencing adapters. Sample-specific IonXPress barcodes added to DNA fragments by ligation. Resulting libraries were quantified with the Ion Library TaqMan[™] Quantitation Kit (Thermo Fisher Scientific, Waltham, USA). Samples were re-amplified by with 10 PCR cycles followed by re-quantification as described above. Samples were mixed at equimolar ratios and the resulting sequencing library pool was subjected to sequencing using the IonChef[™] in combination with the IonTorrent[™] Genestudio S5 Plus sequencer (Thermo Fisher Scientific, Waltham, USA) and the Ion 550 Chip Kit. Samples were sequenced to a mean sequencing depth of 6 million reads per samples. Raw sequencing counts were quality filtered using Trimmomatic 0.39 with a read length cutoff of 100 bp and a sliding window of 10 with a quality value of 15. Sequencing adapters were removed with cutadapt 3.4. Human sequences were removed with bowtie2 by mapping to the GRCh38 (hg38) genome. MetaPhlAn4⁸³ and the ChocoPhlAn v296 pangenome database were used to assign microbial taxonomy to sequences. The HMP Unified Metabolic Analysis Network 3 (HUMAN3⁸⁴) was used to predict functional information by mapping to the UniRef90 database version 201901b. Predicted gene families were mapped to the MetaCyc⁸⁵ database for pathway-based and the KEGG⁸⁶ database for gene-family level analysis respectively (Supp. Table 5). Maaslin2⁸³ was used to identify significantly differential abundant metabolic (Humann3) or taxonomic (Metaphlan4) features from log-transformed relative abundances. The intestinal immunomodulatory risk index or MOFA factor levels 1 and 3 were used as fixed and individual patients as random effects for the linear model applying an alpha value of 0.25 for the Benjamini & Hochberg-procedure corrected p-values. Relative abundances were summed for MetaCyc superclasses for all pathways comprising them (Supp. Table 5).

Virome Analysis

Extraction of nucleic acids: Stool samples were mixed with 10 ml of filtered phage SM buffer and vortexed vigorously for 4 h at 4 °C, then centrifuged at 4000 g for 30 min to collect supernatant. The supernatant was passed through 0.22 µm filters (PES Membrane,

Lot No. ROCB29300, Merck Millipore, Co., Cork, Ireland) to remove bacterial-associated particles, and the volume was subsequently concentrated to less than 50 μ L by Amicon® Ultra Centrifugal Filters (10 kDA, Lot No. R9EA18187, Merck Millipore, Co., Cork, Ireland). Then 1/5 volume of chloroform was mixed with the samples and centrifuged at 14,000 g for 3 min, retaining the upper phase followed by a DNase I (1 U/ μ L, Lot No. 1158858, Invitrogen, Carlsbad, CA, USA) treatment for 1 h at 37 °C to remove non-phage DNA. DNase I was inactivated by adding EDTA (0.1M). Subsequently, lysis buffer (700 μ L KOH stock (0.43 g/10 mL), 430 μ L DDT stock (0.8 g/10 mL), and 370 μ L water, pH = 12) was added to the reaction and incubated at room temperature for 10 min followed by 2 h incubation at -80 °C, and 5 min at 55 °C. Lysed VLPs were then treated for 30 min at 55 °C with Proteinase K (20 mg/mL, Lot No. 1112907, Invitrogen, Carlsbad, CA, USA) to digest remaining viral capsid and extract the virome DNA. AMPure beads (Agencourt, Beckman Coulter, Brea, CA, USA) were added to the extracted DNA and incubated for 15 min at room temperature. DNA was eluted from beads by 35 μ L Tris buffer (10 mM, pH = 9.8) and stored at -80 °C. Sequencing was performed on an Illumina NovaSeq-PE150 platform.

Virome sequence processing: Virome sequencing data was analyzed using the ViroProfiler pipeline.⁸⁷ Raw reads were processed with fastp (v0.20.1) to remove adaptors and low-quality bases. Remaining reads were deduplicated using dedupe.sh from bbmap suite (v38.76) (<https://sourceforge.net/projects/bbmap>). The obtained reads were assembled into contigs using metaSPAdes (v3.14.0)⁸⁸ with default parameters retaining only contigs longer than 1 kb. Redundant contigs were removed using dedupe.sh. Remaining contigs were used to predict viral sequences by the combination of VirSorter2⁸⁹, CAT (v5.0.4)⁹⁰ and DeepVirFinder (v1.0)⁹¹. Contigs predicted as category 1 and 2 by VirSorter2, or predicted as viruses by CAT, were classified as viruses. Contigs were also classified as viruses if they were predicted as category 3 by VirSorter2 or could not be classified to taxonomy by CAT but predicted as a virus by DeepVirFinder with q value < 0.01. To further reduce the redundancy of the contig library, we clustered contigs following the “rapid genome clustering based on pairwise ANI” protocol in CheckV, which ensures the contigs in the same cluster sharing more than 95% identity and 80% coverage on the short contigs. The longest contig of each cluster were selected as representatives for downstream analysis. Clean reads of each sample were mapped to the representative contigs using minimap2. Abundance profiles including read counts and TPM were calculated using CoverM (v0.6.1) (<https://github.com/wwood/CoverM>). Viral bins were constructed using PHAMB⁹². Abundance of each bin was obtained by aggregate abundance of their contig members. ORFs were predicted using Prodigal (v2.6.3) and provided to vConTACT2⁹³ for taxonomy annotation. For contigs and bins that could not be assigned a taxonomy by vConTACT2, CAT annotations were used.

Data analysis: Viral richness (Ace) and diversity (Inverse Simpson’s diversity index) and Principal Coordinates Analysis (PCoA) that based on “Bray-Curtis” similarities was conducted using vegan package in R (v4.0.2). The significant difference in PCoA was determined by Permutational Multivariate Analysis of Variance (PERMANOVA). Means of groups were compared with either Kruskal-Wallis with Dunn’s post hoc test or the Wilcoxon Rank Sum test (two-sided) adjusted for multiple comparisons using the “Bonferroni Holm”

method. The corr.test function in Psych R package v2.1.9 was used to compute the Pearson correlation and multiple comparisons were corrected using Holm method ($\alpha=0.05$). Graphs were generated in GraphPad Prism version 9 and R ggplot2 package.

BCoAT phylogenetic analysis in virome: *BCoAT* was identified in viral genes using hmmscan searching against HMM profile model of *BCoAT* in NCBI protein family database (accession: TIGR03948.1). There are two viral contigs contain *BCoAT* genes, each of them in a VC cluster that contains more than one member contigs (VC-1 belongs to viral cluster VC_804_0, and VC-2 belongs to VC_1122_0). Homologous genes of bacteriophage-coded *BCoAT* were obtained using MMseqs2⁹⁴ to search against NCBI NR database. Obtained homologous protein sequences were aligned using MAFFT v7.475, trimming with trimAl v1.4.rev15 to remove bases with <50% coverage, and creating a phylogenetic tree using FastTree v2.1.10⁹⁵. The phylogenetic tree was visualized using ggtree package in R.

Viral gene annotation and genome comparison: Viral genome of VC-1 and VC-2 was annotated using the PHROGs database⁹⁶, and visualized by pyCircos (<https://github.com/ponnhide/pyCircos>). The closest related bacterial gene to VC-1 and VC-2 on the *BCoAT* phylogenetic tree was selected as the reference genome. tBLASTx was used to compare it to the viral contig. The gggenomes package in R (<https://github.com/thackl/gggenomes>) was used to visualize the results of the comparison, with lines representing the protein sequence identity determined by tBLASTx.

Metabolome Analysis

Sample preparation for targeted analysis: Approximately 100 mg of stool was weighed in a 15 mL bead beater tube (CKMix50 15 mL, Bertin Technologies, Montigny-le-Bretonneux, France) filled with 2.8 mm and 5.0 mm ceramic beads i.d. 5 mL of methanol-based dehydrocholic acid extraction solvent ($c=1.3 \mu\text{mol/L}$) was added as an internal standard for work-up losses. The samples were extracted with a bead beater (Precellys Evolution, Bertin Technologies) supplied with a Cryolys cooling module 3 times each for 20 seconds with 15 seconds breaks in between at 10,000 rpm. 1 mL of the fecal suspension was dried in a vacuum centrifuge (Eppendorf Vacufuge) to determine the dry weight.

Measurement of SCFA, lactic acid, desaminotyrosine and indole-3-carboxyaldehyde (Panel 1): The 3-NPH method was used for the quantitation of SCFAs as well as lactic acid, desaminotyrosine and indole-3-carboxyaldehyde⁹⁷. Briefly, 40 μL of the fecal extract and 15 μL of 50 μM isotopically labeled standards were mixed with 20 μL 120 mM EDC HCl-6% pyridine-solution and 20 μL of 200 mM 3-NPH HCL solution. After 30 min at 40°C and shaking at 1000 rpm using an Eppendorf Thermomix (Eppendorf, Hamburg, Germany), 900 μL acetonitrile/water (50/50, v/v) was added. After centrifugation at 13000 U/min for 2 min the clear supernatant was used for analysis. The measurement was performed using a QTRAP 5500 triple quadrupole mass spectrometer (Sciex, Darmstadt, Germany) coupled to an ExionLC AD (Sciex, Darmstadt, Germany) ultrahigh performance liquid chromatography system. The electrospray voltage was -4500 V, curtain gas 35 psi, ion source gas 1 55 psi, ion source gas 2 65 psi and the temperature 500°C. The MRM-parameters were optimized using commercially available standards. The

chromatographic separation was performed on a 1.7 μm , 100 $\times$ 2.1 mm, 100 Å Kinetex C18 column (Phenomenex, Aschaffenburg, Germany) column with 0.1% formic acid (eluent A) and 0.1% formic acid in acetonitrile (eluent B) as elution solvents. An injection volume of 1 μL and a flow rate of 0.4 mL/min was used. The gradient elution started at 23% B which was held for 3 min, afterward the concentration was increased to 30% B at 4 min, with another increase to 40%B at 6.5 min, at 7 min 100% B was used which was hold for 1 min, at 8.5 min the column was equilibrated at starting conditions. The column oven was set to 40°C and the autosampler to 15°C. Data acquisition and instrumental control were performed with Analyst 1.7 software (Sciex, Darmstadt, Germany). The data was analyzed with MultiQuant 3.0.3 (Sciex, Darmstadt, Germany) and Metaboanalyst⁹⁸. Features with more than 70% of missing values were removed. Missing values were replaced by LoDs (1/5 of the minimum positive value of each variable). Heatmaps were generated after normalization by log transformation (base 10). Means of multiple groups (time-points) were compared with the Kruskal-Wallis test followed by Dunn's multiple comparisons test. Comparisons between two groups (No ABX vs ABX; No GI-GvHD vs GI-GvHD) were performed with the Wilcoxon matched-pairs signed rank test or the Mann-Whitney test, respectively (two-sided, 95% confidence level). Graphs and statistics were created in Prism 9 (GraphPad).

Measurement of bile acids (Panel 2): Bile acid analysis was performed according to Reiter et al. 2020⁹⁹. Briefly, 20 μL of isotopically labeled BAs (ca. 7 μM each) were added to 100 μL of fecal extract. Targeted bile acid measurement was performed using the same system as described above. A multiple reaction monitoring (MRM) method was used for the detection and quantification of the BAs⁹⁹. An electrospray ion voltage of -4500 V and the following ion source parameters were used: curtain gas (35 psi), temperature (450 °C), gas 1 (55 psi), gas 2 (65 psi), and entrance potential (-10 V). The MS parameters and LC conditions were optimized using commercially available standards of endogenous BAs and deuterated BAs, for the simultaneous quantification of selected 34 analytes. For separation of the analytes a 100 $\times$ 2.1 mm, 100 Å, 1.7 μm , Kinetex C18 column (Phenomenex, Aschaffenburg, Germany) was used. Chromatographic separation was performed with a constant flow rate of 0.4 mL/min using a mobile phase consisted of water (eluent A) and acetonitrile/water (95/5, v/v, eluent B), both containing 5 mM ammonium acetate and 0.1% formic acid. The gradient elution started with 25% B for 2 min, increased at 3.5 min to 27% B, in 2 min to 35% B, which was hold until 10 min, increased in 1 min to 43% B, held for 1 min, increased in 2 min to 58% B, held 3 min isocratically at 58% B, then the concentration was increased to 65% at 17.5 min, with another increase to 80% B at 18 min, following an increase at 19 min to 100% B which was hold for 1 min, at 20.5 min the column was equilibrated for 4.5 min at starting. The injection volume for all samples was 1 μL , the column oven temperature was set to 40 °C, and the auto-sampler was kept at 15 °C. Data acquisition and instrumental control were performed with Analyst 1.7 software (Sciex, Darmstadt, Germany). The data was analyzed as indicated in Panel 1.

Multi-omics Factor Analysis (MOFA): Details on MOFA methodology and downstream analysis can be found in the Supplementary Notes and the original publications¹⁰⁰.

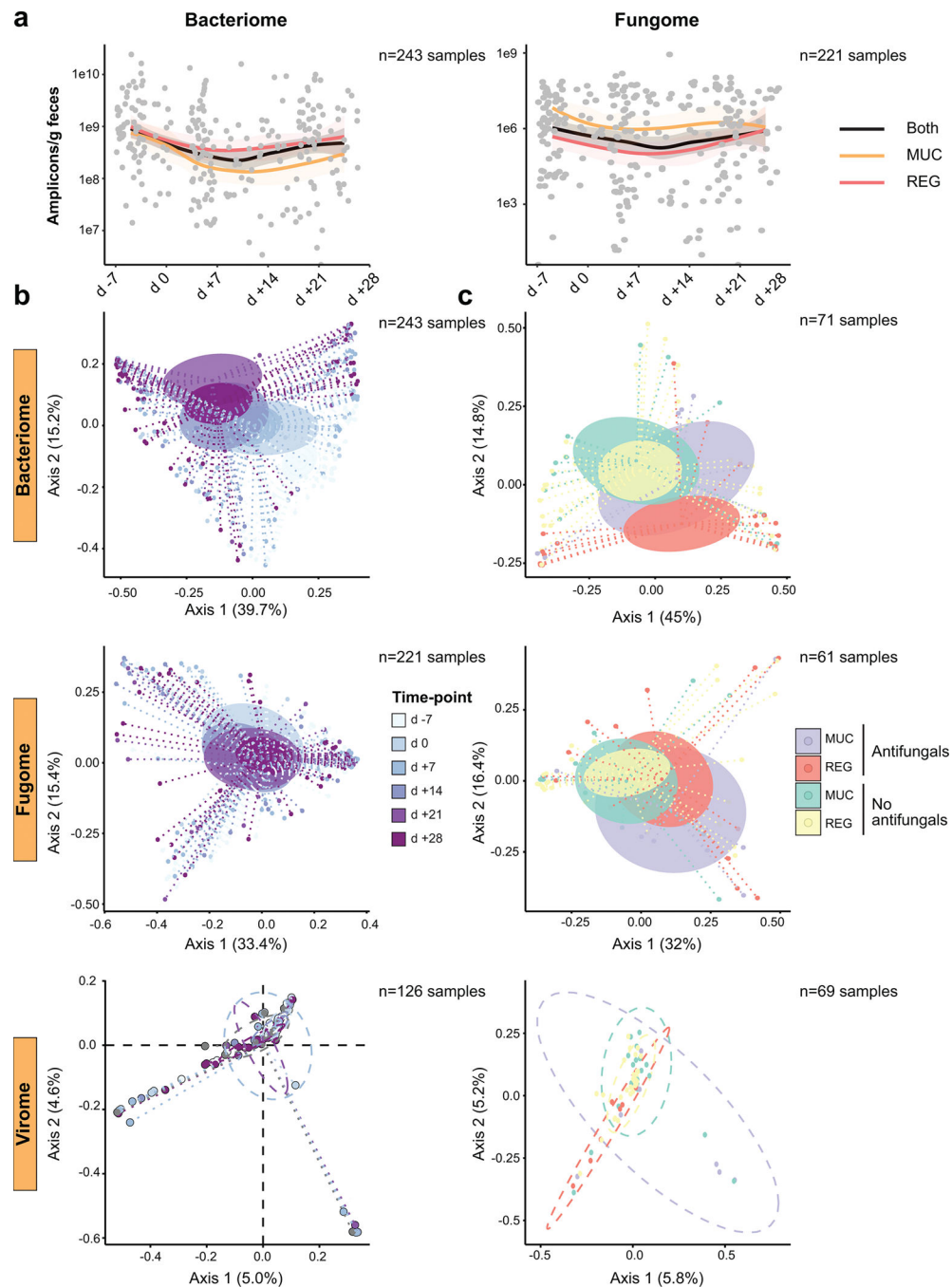
As input for the MOFA model, we used bacterial 16S rRNA (239 features) and fungal ITS1 rRNA (154 features) ASVs as well as assembled viral clusters (434 features) and metabolite concentrations from targeted metabolomics (28 features, metabolites with missing values in more than 90% of samples were excluded from the model). Please refer to Supplementary Notes for a description of the pipeline used to generate ASVs for MOFA. All inputs were normalized by centralized log normalization (microbiome/virome data) or log transformation (base 10, metabolomics data). When fitting the model, we selected for the top factors ordered by the mean fractional variance explained across omics modalities (i.e., Factor 1 contributed the most, Factor 10 the least to mean fractional variation, Figure 3B).

When testing Factor values for statistically significant differences between groups (clinical metadata, IMM-RI and *BCoAT* data), a two-tailed Mann-Whitney-U test was used. The mean of Factor values of days +7–21 post allo-SCT of a given patient were used for testing. In cases where different group assignment was possible for the same patient for different timepoints (e.g., *BCoAT* gene copies detected at day 7, but not at day 14), the patient was assigned to the group showing the characteristic, if the characteristic was detected for at least one timepoint. The resulting p-values were corrected for multiple testing using the FDR approach¹⁰¹ across all hypotheses where Factor values were analyzed for association with clinical metadata, IMM-RI or *BCoAT* data.

To assess time-dependent variations, we fitted a MEFISTO²⁵ model on the MOFA input data, with time-points in days relative to allo-SCT as the covariate. Due to sparseness, time-points after day 35 were combined to a single timepoint.

MOFA/MEFISTO was performed using R (4.1.1) and Python (3.6.13). The MOFA/MEFISTO model was trained using the MOFA2 R-package (1.4.0)¹⁰⁰ as an interface to the mofapy2 Python-package (0.6.4). Scale differences between views were not pronounced, therefore MOFA input modalities were not scaled to unit variance. The number of Factors was set to 10, and convergence was set to medium. Downstream analysis of MOFA, statistical testing and plotting were done in R, using the packages MOFA2 (1.4.0), ggplot2 (3.3.5), ggpubr (0.4.0), corrplot (0.91) and rstatix (0.7.0).

Extended Data

**Extended Data Fig. 1.**

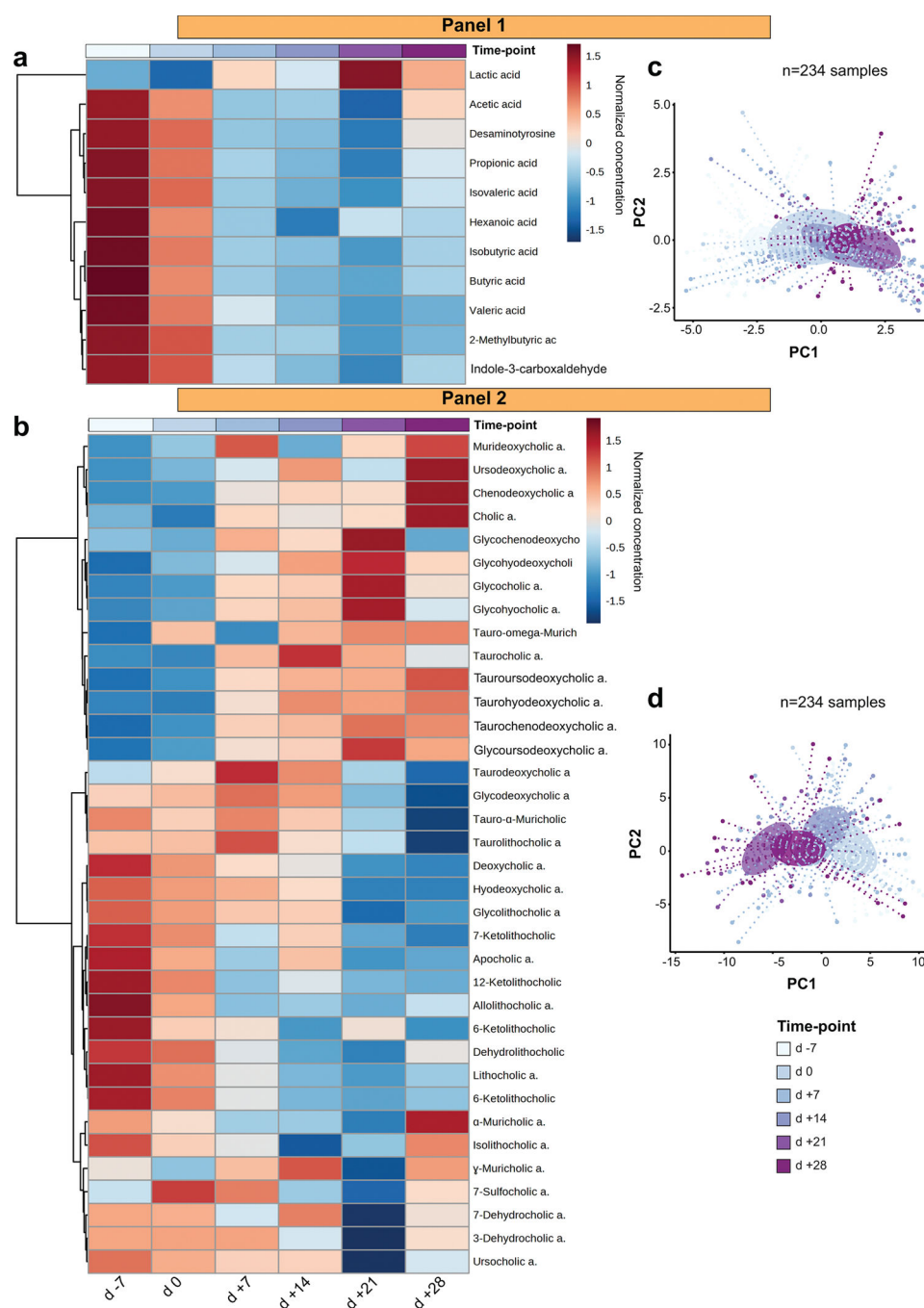
Longitudinal dynamics of bacterial, fungal and viral community compositions in patients receiving allo-SCT.

a) Quantification of the bacterial and fungal load by 16S and 28S rDNA copy numbers per gram of stool at time-points relative to allo-SCT (Day 0). Solid lines represent the smoothed conditional means for the entire cohort (black), Munich (MUC, orange) or Regensburg (REG, red) calculated by locally weighted regression using the locally estimated scatterplot

smoothing (LOESS) method. The gray shading indicates the 95% confidence interval for entire cohort. Each individual patient stool sample is plotted as a gray dot superimposed on the graph. The number of samples is indicated.

b) Beta diversity analysis illustrating changes in bacteriome, fungome and virome by time-points relative to allo-SCT (Day 0). For bacteriome and fungome, beta diversity was calculated by weighted UniFrac. For virome, beta diversity was calculated by Bray-Curtis dissimilarity. Distances were projected in Principal coordinate analysis (PCoA). Bacteriome: Comparisons between Day +7 ($p=0.001$), Day + 14 ($p=0.001$), Day +21 ($p=0.001$), Day +28 ($p=0.001$) vs baseline (Day -7) are significant (pairwise Adonis test adjusted for multiple comparisons). Virome: Comparisons between Day +7 ($p=0.013$) and Day +28 ($p=0.001$) vs baseline are significant (PERMANOVA test). The number of samples is indicated.

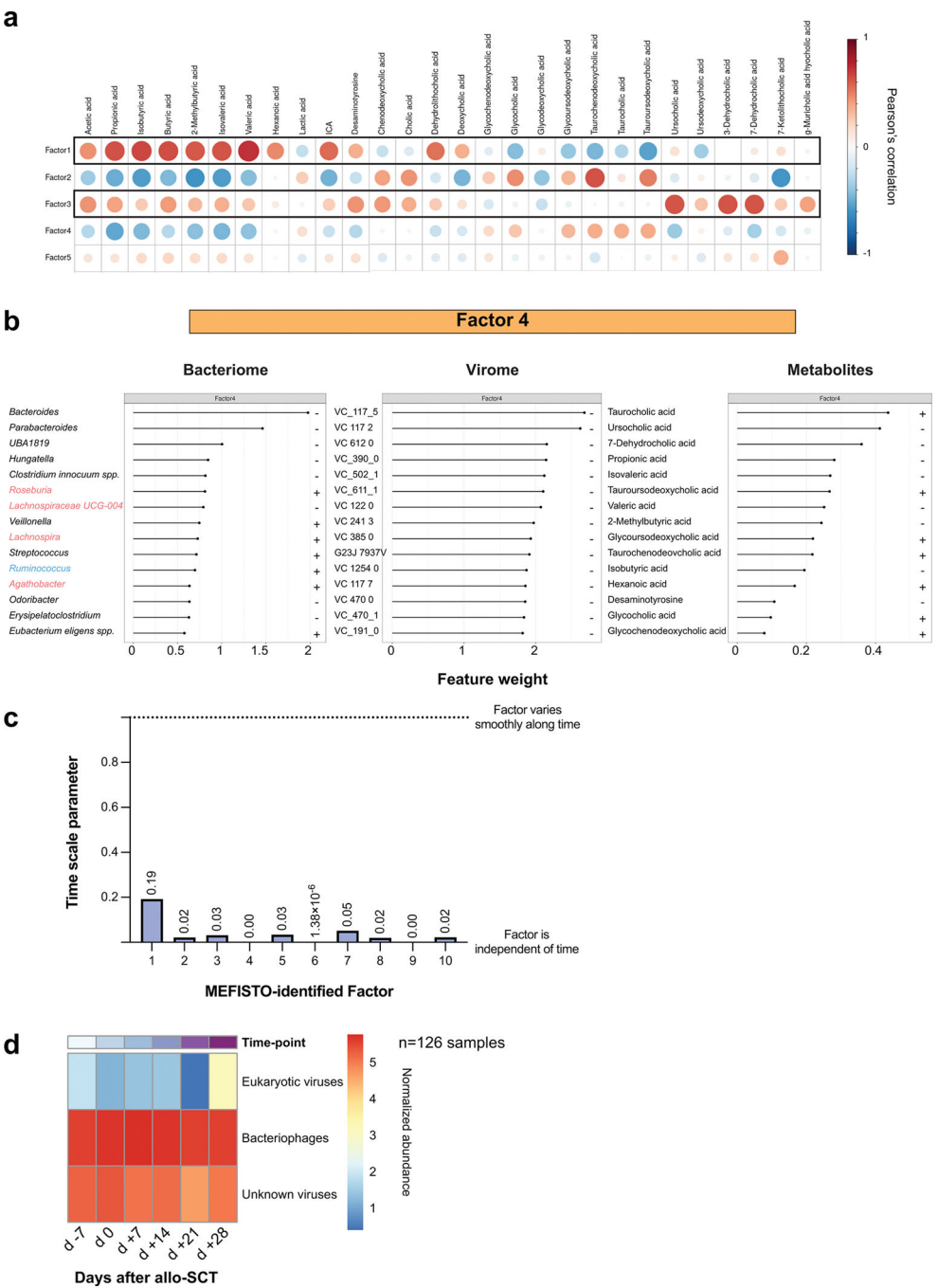
c) Beta diversity analysis illustrating changes in bacteriome, fungome and virome according to study center (MUC or REG) and whether patients received antifungal therapy ("No Antifungals" or "Antifungals"). For bacteriome and virome, beta diversity was calculated by weighted UniFrac. For virome, beta diversity was calculated by Bray-Curtis dissimilarity. Distances were projected in Principal coordinate analysis (PCoA). The number of samples is indicated.

**Extended Data Fig. 2.**

Longitudinal dynamics of SCFAs, BCFAs, IIMs, primary and secondary bile acids in patients receiving in allo-SCT

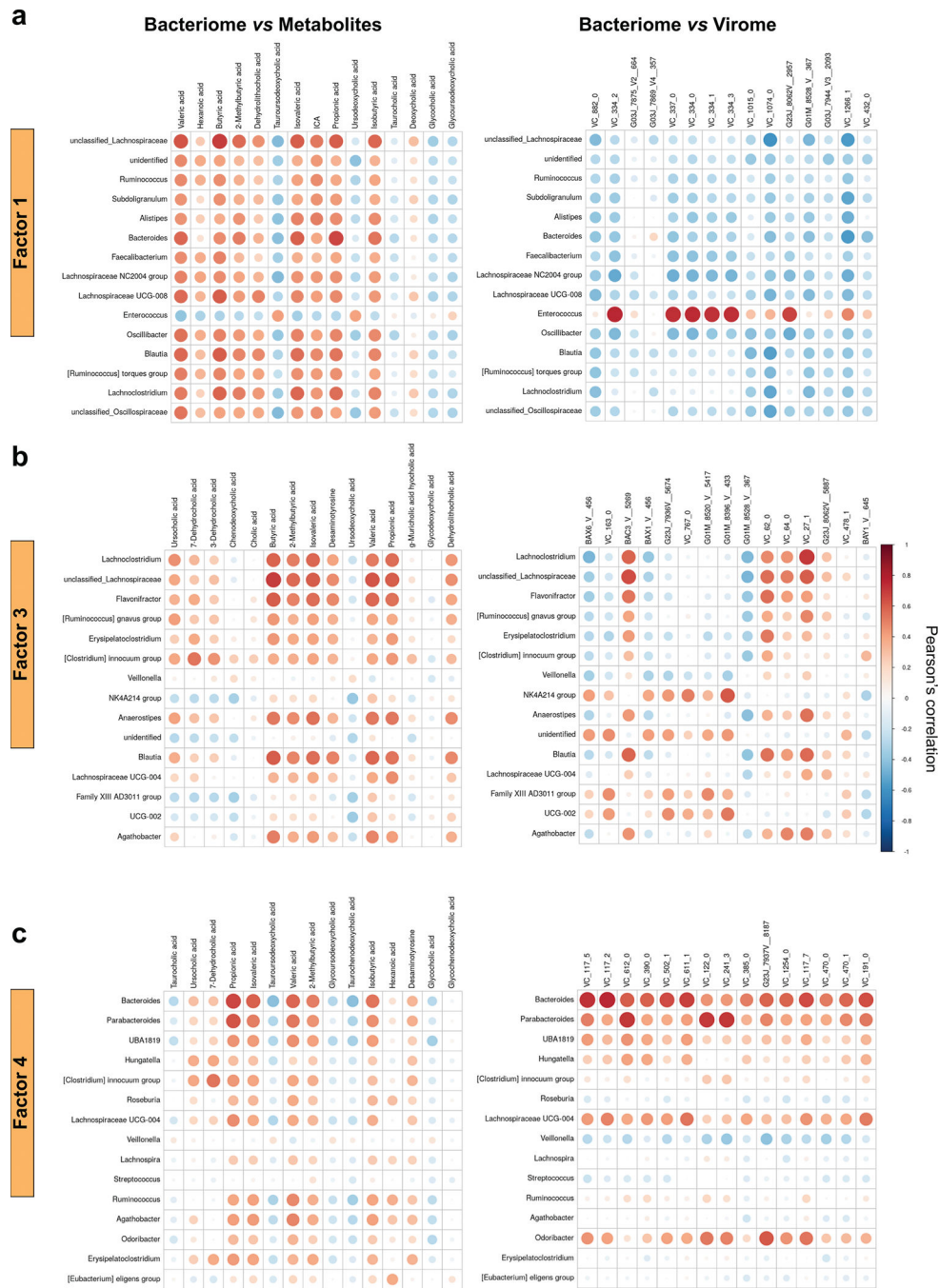
a) Heatmap of normalized Panel 1 metabolite levels in stool samples of allo-SCT patients averaged by time-points relative to allo-SCT. Normalized concentrations are indicated in the adjacent color legend. Clustering based on metabolite expression patterns using the Ward algorithm. Distance measure is Euclidian. The number of samples is indicated.

- b) Heatmap of normalized Panel 2 metabolite levels in stool samples of allo-SCT patients averaged by time-points relative to allo-SCT. Normalized concentrations are indicated in the adjacent color legend. Clustering based on metabolite expression patterns using the Ward algorithm. Distance measure is Euclidian. The number of samples is indicated.
- c) Principal component analysis (PCA) of Panel 1 metabolite profiles by time-points relative to allo-SCT. Comparisons at Day 0 ($p=0.004$), Day +7 ($p=0.001$), Day +14 ($p=0.001$), Day +21 ($p=0.001$) and Day +28 ($p=0.001$) vs Day -7 are significant (pairwise Adonis test of Euclidean distances adjusted for multiple comparisons).
- d) PCA of Panel 2 metabolite profiles by time-points relative to allo-SCT. Comparisons at Day +7 ($p=0.001$), Day +14 ($p=0.001$), Day +21 ($p=0.001$) and Day +28 ($p=0.001$) vs Day -7 are significant (pairwise Adonis test of Euclidean distances adjusted for multiple comparisons).



Extended Data Fig. 3.
MOFA (multi-omics factor analysis) and MEFISTO (a method for the functional integration of spatial omics data) in allo-SCT patients
a) Correlation between MOFA-identified Factors and normalized intestinal metabolite concentrations. Associations between Factor values and metabolites were analyzed via Pearson's correlation. The correlation coefficient is indicated in the adjacent color legend. The p-values associated with the correlations were corrected for multiple testing with the FDR approach.

- b) Top 15 Features in bacteriome, virome and metabolites in Factor 4 in descending order according to Feature weight. Larger weights indicate a higher correlation with that Factor, while the positive or negative sign indicates the directionality of that variation, i.e., “+” indicates a positive association, “-“ a negative association.
- c) Bar plot of time scale parameters assigned to Factors 1 through 10 identified by MEFISTO. MEFISTO assigns a time scale value between 0 and 1 to each Factor, which reflects the degree to which that Factor is dependent on time. A value of 0 implies no time-dependency, a value of 1 strong time-dependency. Of note, results pertaining to the identified Factors, their weight/covariance structure, variance explained across omics entities and Factor values obtained by MEFISTO modelling were almost identical and thus comparable to the output of our original MOFA model.
- d) Heatmap of normalized abundance of viral contigs assigned to eukaryotic and prokaryotic viruses at time-points relative to allo-SCT. The number of samples is indicated.

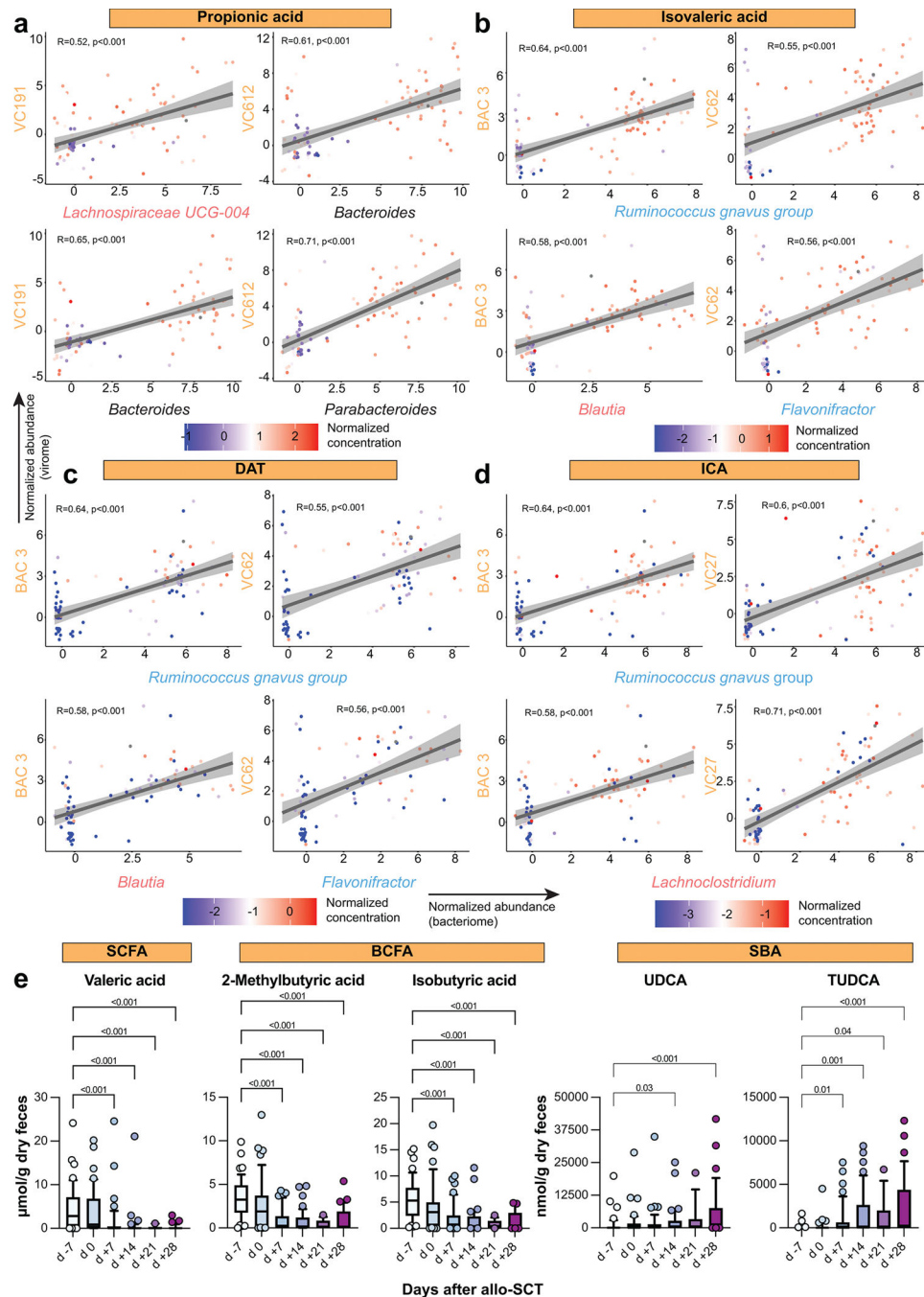


Extended Data Fig. 4.
Correlation between top 15 bacterial and metabolite as well as bacterial and viral high-weight Features group in Factors 1, 3 and 4.
a) Heatmaps of pairwise Pearson's correlations of top Features across different omics modalities. The Feature values of the top 15 high-weight Features of a given omics modality were correlated with that of another omics modality. (Left) bacterial taxa at genus level and metabolites, (right) bacterial taxa at genus level and bacteriophages at species level. The correlation coefficient is indicated in the adjacent color legend. The p-values associated with

Pearson correlation have been corrected for multiple testing by applying the FDR approach to each set of correlations of two omics modalities of a given Factor.

b) As in a) for Factor 3.

c) As in a) for Factor 4.



Extended Data Fig. 5.

Co-abundance of MOFA-identified bacterial and viral Features is associated with high-level IMM expression, which declines progressively after allo-SCT.

a) Correlation scatter plots of high-weight Features within Factor 4, comparing normalized abundance of bacterial taxa at genus level (x-axis) with that of bacteriophages at species level (y-axis) together with the metabolite propionic acid. Dots represent samples from individual patients at different time points (91 samples from 45 patients), all of which had both 16S and viral metagenomic sequencing data. Dots are colored by intestinal levels of propionic acid, or in grey if no propionic acid data was available. Normalized concentrations of metabolites are indicated in the adjacent heatmap. Associations between bacterial genera and viral species were analyzed via Pearson correlation and linear regression. The R- and p-values are indicated in each plot. The regression line is drawn in blue and the 95 % confidence interval of the regression line is shaded in grey.

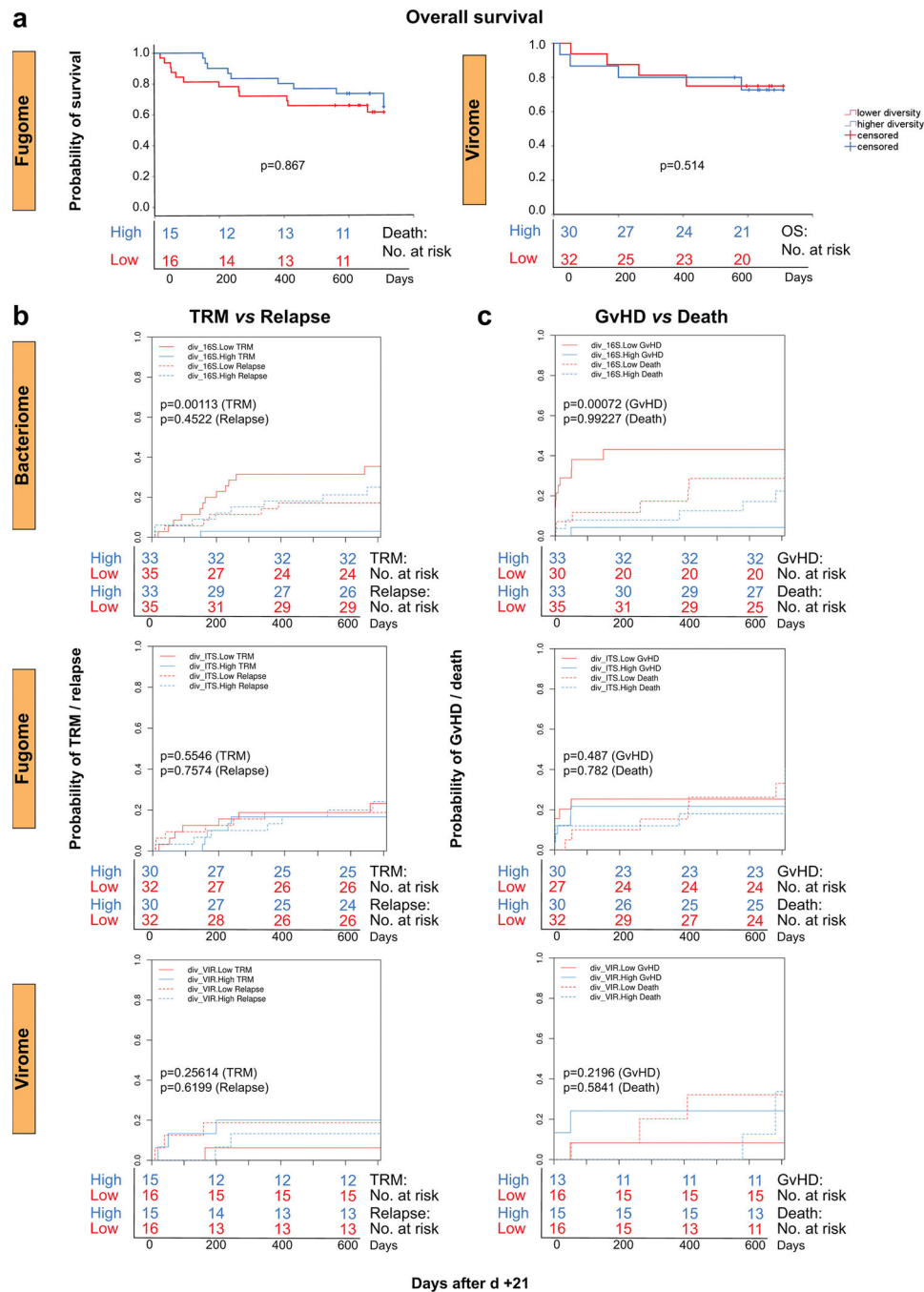
b) As in a) for Features in Factor 3 and the metabolite isovaleric acid.

c) As in a) for Features in Factor 3 and the metabolite DAT.

d) As in a) for Features in Factor 3 and the metabolite ICA.

e) Levels of intestinal microbiota-derived metabolites at time-points relative to allo-SCT (Day 0) in μmol per gram of dried stool measured by targeted mass spectrometry. Number of patients per time-point is indicated in Figure 4D. Significance by two-sided Kruskal-Wallis test corrected for multiple testing via Dunn's test of all time-points against baseline (Day -7).

In the box plots, the box ranges from the 25th to 75th percentiles. The line in the middle of the box is plotted at the median. The whiskers are drawn down to the 10th and up to the 90th percentile. Points below and above the whiskers are drawn as individual points. Points indicate individual patient stool samples sampled at the specified time-points. SBA... secondary bile acids; UDCA...ursodeoxycholic acid; TUDCA...tauroursodeoxycholic acid.



Extended Data Fig. 6.

Intestinal bacterial but not fungal nor viral diversity predicts outcome after allo-SCT

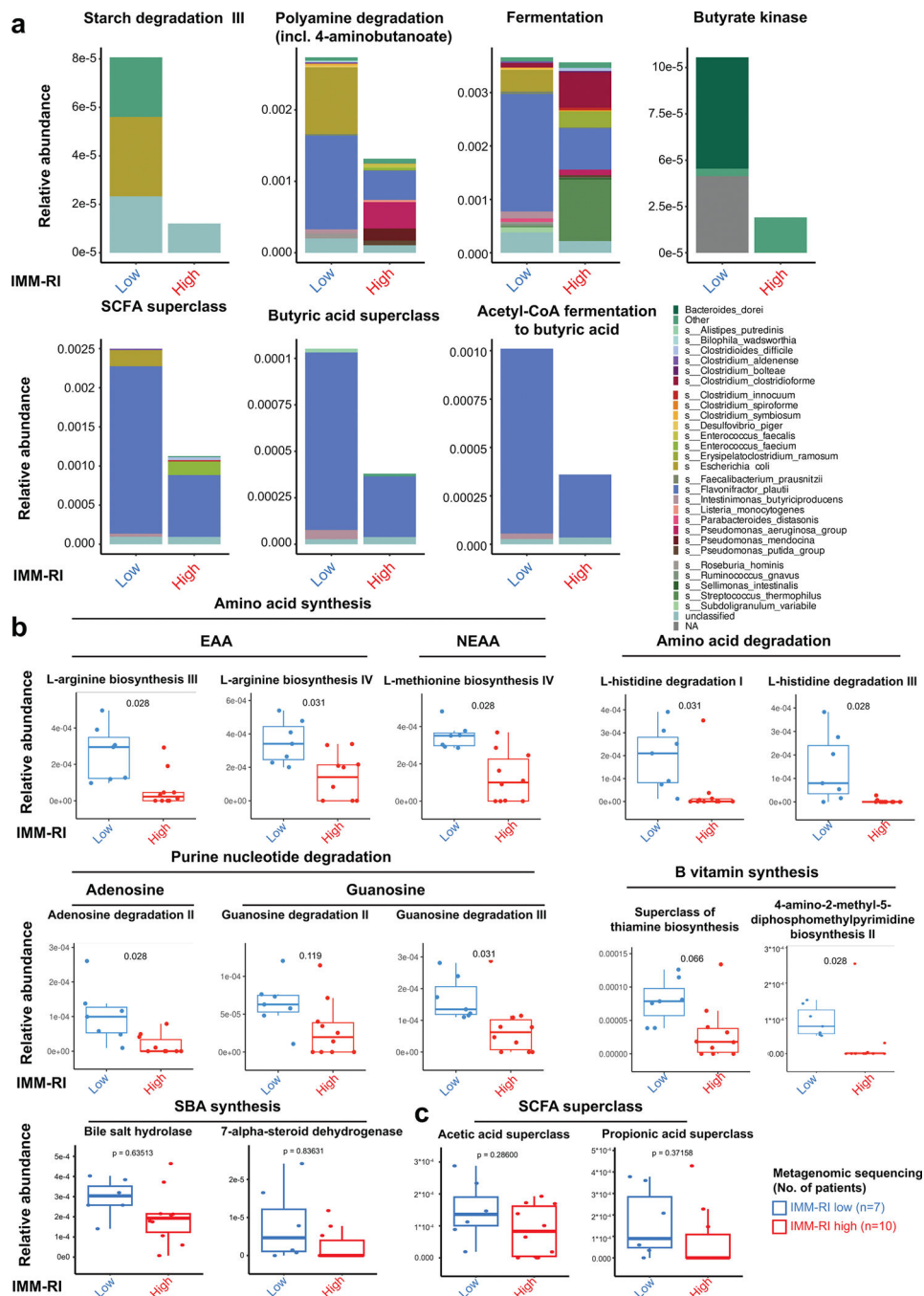
a) 2-year overall survival after Day 21 stratified according to higher and lower fungal (left) and viral (right) alpha diversity. The mean alpha diversity of patient samples at Days +7–21 was calculated and patients were stratified into higher (blue curve) and lower (red curve) diversity groups, defined as above or below the center-specific median Inverse Simpson's diversity index. For fungome, there were 12 deaths among 32 patients in the lower-diversity group (estimated mean survival time 521 (95% CI 428–614) days) and 10 deaths among 30

patients in the higher-diversity group (estimated mean survival time 597 (95% CI 522–672) days). For virome, there were 4 deaths among 16 patients in the lower-diversity group (estimated mean survival time 587 (95% CI 479–696) days) and 4 deaths among 15 patients in the higher-diversity group (estimated mean survival time 576 (95% CI 450–701) days).

Analysis via Kaplan–Meier estimator, significance according to the log-rank test.

b) 2-year cumulative incidence of relapse and transplantation-related mortality (TRM) in a competing risks analysis stratified according to higher and lower bacterial (top), fungal (middle) and viral (bottom) alpha diversity, calculated as in a). For bacteriome, there were 12 cases of TRM and 6 relapses among 35 patients in the lower-diversity group and one case of TRM and 8 relapses among 33 patients in the higher-diversity group. For fungome, there were 7 cases of TRM and 6 relapses among 32 patients in the lower-diversity group and 5 cases of TRM and 7 relapses among 30 patients in the lower-diversity. For virome, there was one case of TRM and 3 relapses among 16 patients in the lower-diversity group and 3 cases of TRM and 2 relapses among 15 patients in the higher-diversity. Significance according to Gray's test.

c) 2-year cumulative incidence of GvHD and its competing risk Death in a competing risks analysis stratified by alpha diversity as in a). For bacteriome, there were 12 cases of GvHD and 6 deaths among 35 patients in the lower-diversity group and 1 case of GvHD and 8 deaths among 33 patients in the higher-diversity group. For fungome, there were 7 cases of GvHD and 6 deaths among 32 patients in the lower-diversity group and 5 cases of GvHD and 7 deaths among 30 patients in the lower-diversity group. For virome, there was one case of GvHD and 3 deaths among 16 patients in the lower-diversity group and 3 cases of GvHD and 2 deaths among 15 patients in the higher-diversity group. Statistics as in b).



Extended Data Fig. 7.

Characterization of differentially abundant microbial pathways and the species which encode them in IMM-RI low- vs high-risk patients via whole shotgun metagenomic sequencing

a) Species-level association with MetaCyc pathways differentially abundant between IMM-RI low and high-risk patients (as shown in Figure 6). The relative abundance and species-level identity of taxa encoding the indicated MetaCyc pathways are shown.

b) Box plot of relative abundance of indicated MetaCyc pathways in IMM-RI low- vs high-risk patients. In the box plots, the box ranges from the 25th to 75th percentiles. The line in the middle of the box is plotted at the mean. The whiskers are drawn down to the minimum and up to the maximum. Samples outside the 1.5-fold IQR were regarded as outliers. Patient samples are plotted as a point superimposed on the graph (high IMM-RI: n=10 patients; low IMM-RI: n=7 patients). Significance by 1-sided Wilcoxon Rank Sum Test corrected for multiple comparisons via the Benjamini & Hochberg correction for multiple testing.

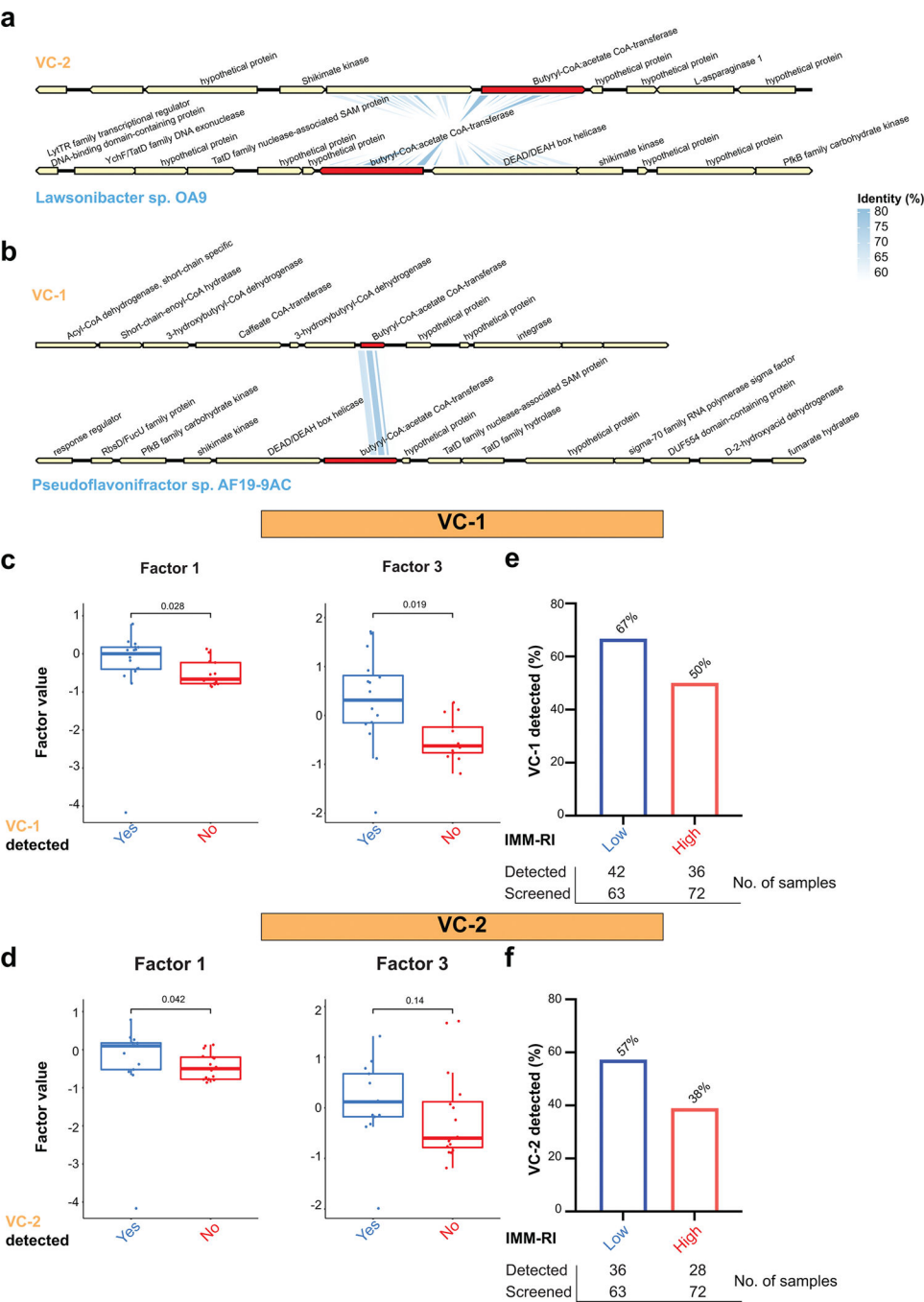
c) Box plot of relative abundance of acetic acid and propionic acid superclasses in IMM-RI low- vs high-risk patients. Plots, numbers and statistics as in b).

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Extended Data Fig. 8.
BCoAT-coding VC-1 and VC-2 bacteriophages are associated with MOFA Factors 1 and 3 and the IMM-RI

a) Gene alignment plot of VC-2. The identity overlap (in percent) is indicated in the adjacent color legend. The *BCoAT* AMG is highlighted in red.

b) Gene alignment plot of VC-1 as in a).

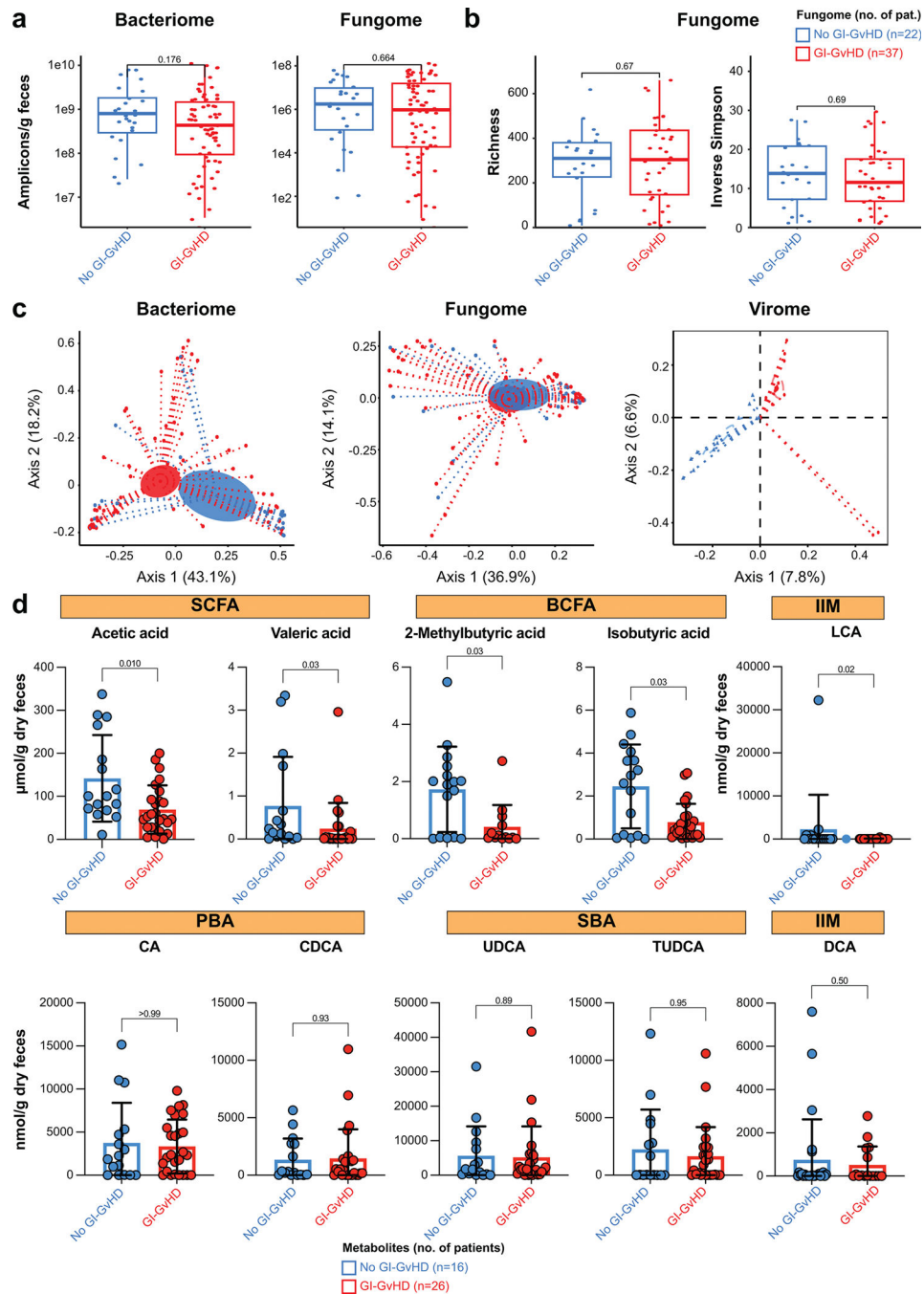
c) Box plots of Factor values for Factors 1 and 3 (averaged across time-points Days +7–21) according to whether VC-1 was detected by viral metagenomic sequencing (“Yes”) or not

(„No“). The center line corresponds to the median, the box ranges from the 25th to the 75th percentile. Whisker length corresponds to the largest/lowest data point that does not exceed the 75th/25th percentile $\pm$ 1.5-fold IQR. Blue: detected (n=16 patients); red: not detected (n=13 patients). Significance by two-tailed Mann-Whitney-U-test corrected for multiple testing via FDR.

d) As in c) for VC-2. Blue: detected (n=13 patients); red: not detected (n=16 patients).

e) Detection of the *BCoAT*-coding VC-1 in patient samples stratified according to IMM-RI. Bar plots show percentage, exact values are provided. The numbers of samples screened vs those in which *VC-1* was detected is indicated below.

f) As in e) for VC-2.

**Extended Data Fig. 9.**

Onset of acute GI-GvHD shifts intestinal bacterial and viral communities and impacts IMM expression profiles

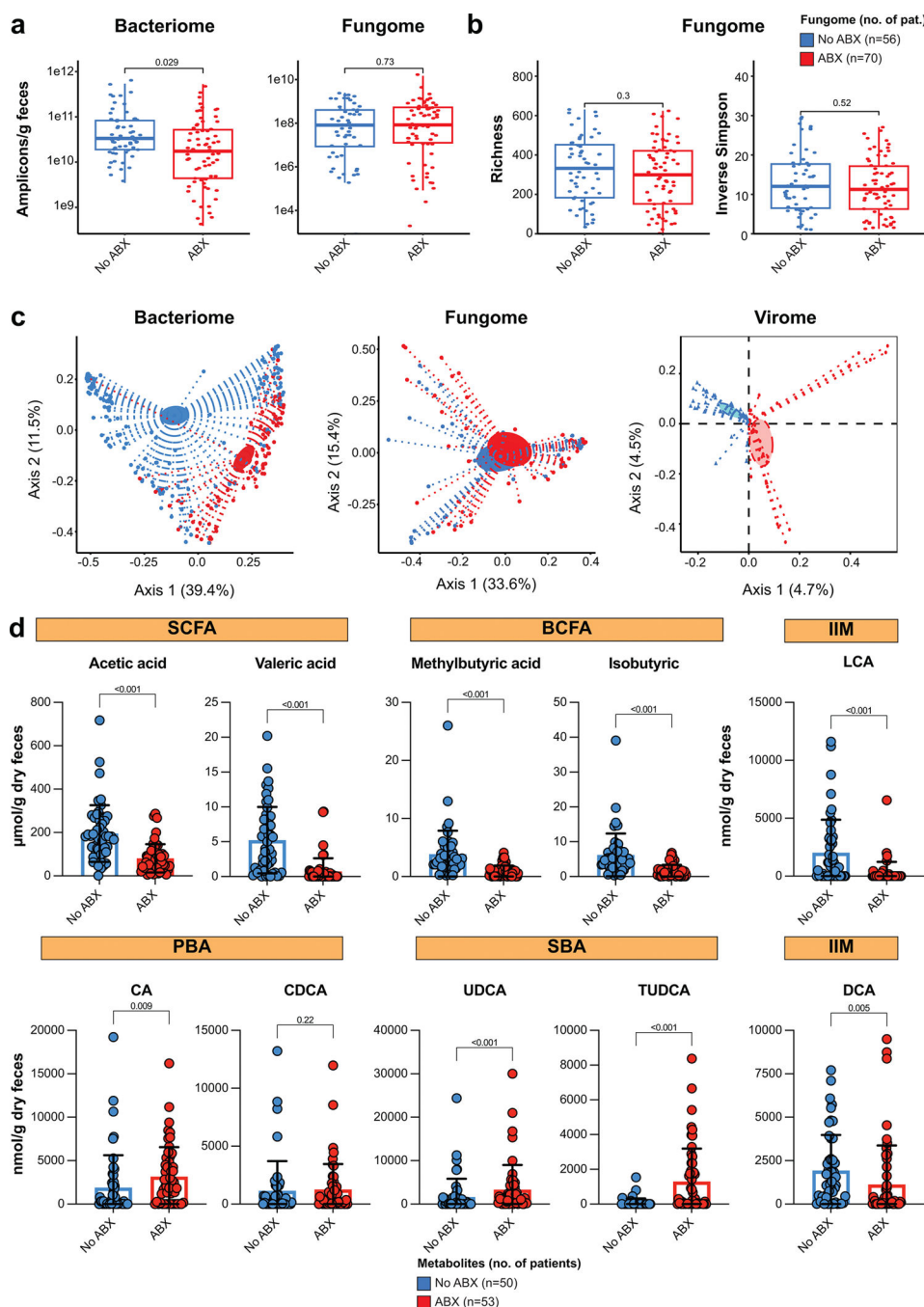
a) Quantification of the bacterial and fungal load by 16S and 28S rDNA copy numbers per gram of stool stratified by patients diagnosed with GI-GvHD (GI-GvHD, red) vs control allo-SCT patients (No GI-GvHD, blue). The box ranges from the 25th to 75th percentiles. The line in the middle of the box is plotted at the median. The whiskers are drawn down to the minimum and maximum. Samples outside the 1.5-fold IQR were regarded as outliers.

Significance by two-tailed Wilcoxon rank sum test adjusted for multiple comparisons via the Benjamini & Hochberg procedure. Each individual patient is plotted as a point superimposed on the graph. Number of patients: for 16S & 28S $n=22$ vs $n=37$, corresponding to „No GI-GvHD“ vs „GvHD“, respectively.

b) Intestinal fungal alpha diversity (Richness, Inverse Simpson's diversity index) stratified by patients diagnosed with GI-GvHD (GI-GvHD) vs control allo-SCT patients (No GI-GvHD). Plots, numbers and statistics as in a).

c) Beta diversity analysis illustrating changes in bacteriome, fungome and virome in patients with GI-GvHD vs control allo-SCT patients. For bacteriome and fungome, beta diversity was calculated by weighted UniFrac. For virome, beta diversity was calculated by Bray-Curtis dissimilarity. Distances were projected in PCoA. Bacteriome: Comparison between patients with GI-GvHD vs No GI-GvHD is significant ($p=0.026$ by one-side pairwise Adonis test). Virome: Comparison between patients with GI-GvHD vs No GI-GvHD is significant ($p=0.03$ by PERMANOVA test).

d) Levels of indicated microbiota-derived metabolites stratified by patients diagnosed with GI-GvHD vs control allo-SCT patients. Significance by two-tailed Mann-Whitney test. In the scatter-dot plots, the box is plotted at the mean. Error bars indicate standard deviation. Each individual patient is plotted as a point superimposed on the graph. The number of patients per group are indicated in the legend.



Extended Data Fig. 10.

Impact of antibiotics on bacterial abundance, bacterial and viral community composition and IMM expression profiles

a) Quantification of the bacterial and fungal load by 16S and 28S rDNA copy numbers per gram of stool stratified by antibiotic exposure: No Antibiotics (“No ABX”, blue) (blue) vs Antibiotics (“ABX”, red). Once a patient was treated with antibiotics, the current and all subsequent samples were classified as “ABX”. The box ranges from the 25th to 75th percentiles. The line in the middle of the box is plotted at the median. The whiskers

are drawn down to the minimum and maximum. Samples outside the 1.5-fold IQR were regarded as outliers. Significance by two-tailed Wilcoxon rank sum test adjusted for multiple comparisons via the Benjamini & Hochberg procedure. Each individual patient is plotted as a point superimposed on the graph. Number of patients: for 16S $n=59$ vs $n=70$, for 28S $n=56$ vs $n=70$, corresponding to „No ABX“ vs „ABX“, respectively.

b) Intestinal fungal alpha diversity (Richness, Inverse Simpson's diversity index) in paired patient samples according to antibiotic status as in a). Plots, numbers and statistics as in a).

c) Beta diversity analysis illustrating the impact of antibiotics on the intestinal bacterial, fungal and viral communities. Each point represents individual patient samples annotated with metadata regarding concomitant antibiotic therapy. For bacteriome and virome, beta diversity was calculated by weighted UniFrac. For virome, beta diversity was calculated by Bray-Curtis dissimilarity. Distances were projected in PCoA. Bacteriome: Comparison between patients with “No ABX” vs “ABX” is significant ($p=0.001$ by one-side pairwise Adonis test). Virome: Comparison between patients with “No ABX” vs “ABX” is significant ($p=0.003$ by PERMANOVA test).

d) Levels of indicated microbiota-derived metabolites in paired patient samples before and after exposure to ABX. Significance by two-tailed Wilcoxon matched-pairs signed rank test. In the scatter-dot plots, the box is plotted at the mean. Error bars indicate standard deviation. Each individual patient is plotted as a point superimposed on the graph. The number of patients per group are indicated in the legend.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability

Microbial sequencing (bacterial, and fungal amplicon data and viral metagenomic sequencing data) that support the findings of this study have been deposited at the European Nucleotide Archive under accession number PRJEB53547 (<https://www.ebi.ac.uk/ena/browser/view/PRJEB53547>). Mass spectrometry data have been deposited at Zenodo under accession number 6603017 (<https://zenodo.org/record/6603017>). Both repositories are annotated with clinical metadata.

Source data for all Figures and Extended Data have been provided as Source Data files. All other data supporting the findings of this study are available from the corresponding author on reasonable request.

The stool samples analyzed in this study comprised a unique bio-sample collection and have been expended for the analyses performed in this study. No additional material is available.

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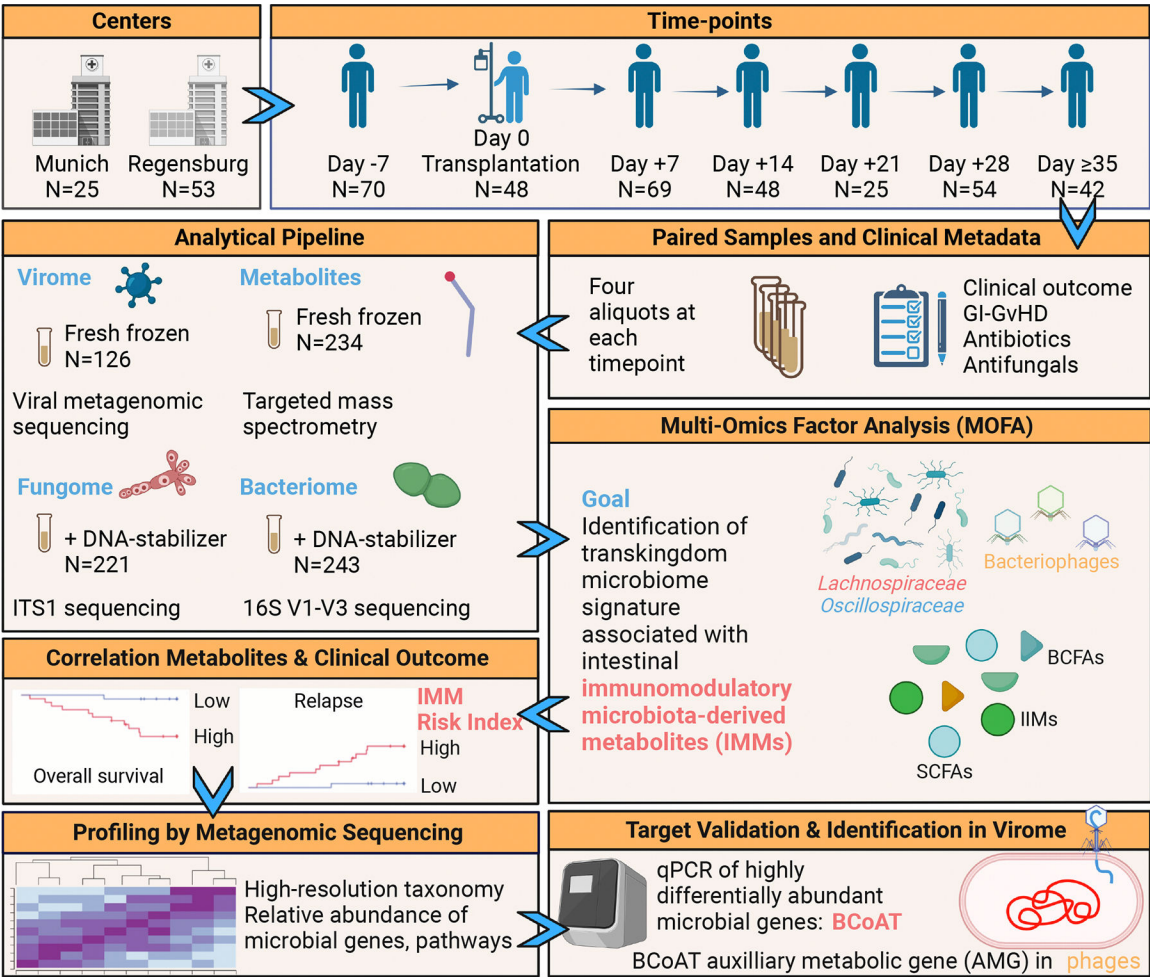


Figure 1:
Study design and sampling scheme
a) Graphical representation of allo-SCT centers including patient numbers per center, sampling time-points including number of patients per time-point, sampling scheme including processing and storage, sample numbers per multi-omics modality, and downstream analytics. Created with BioRender.

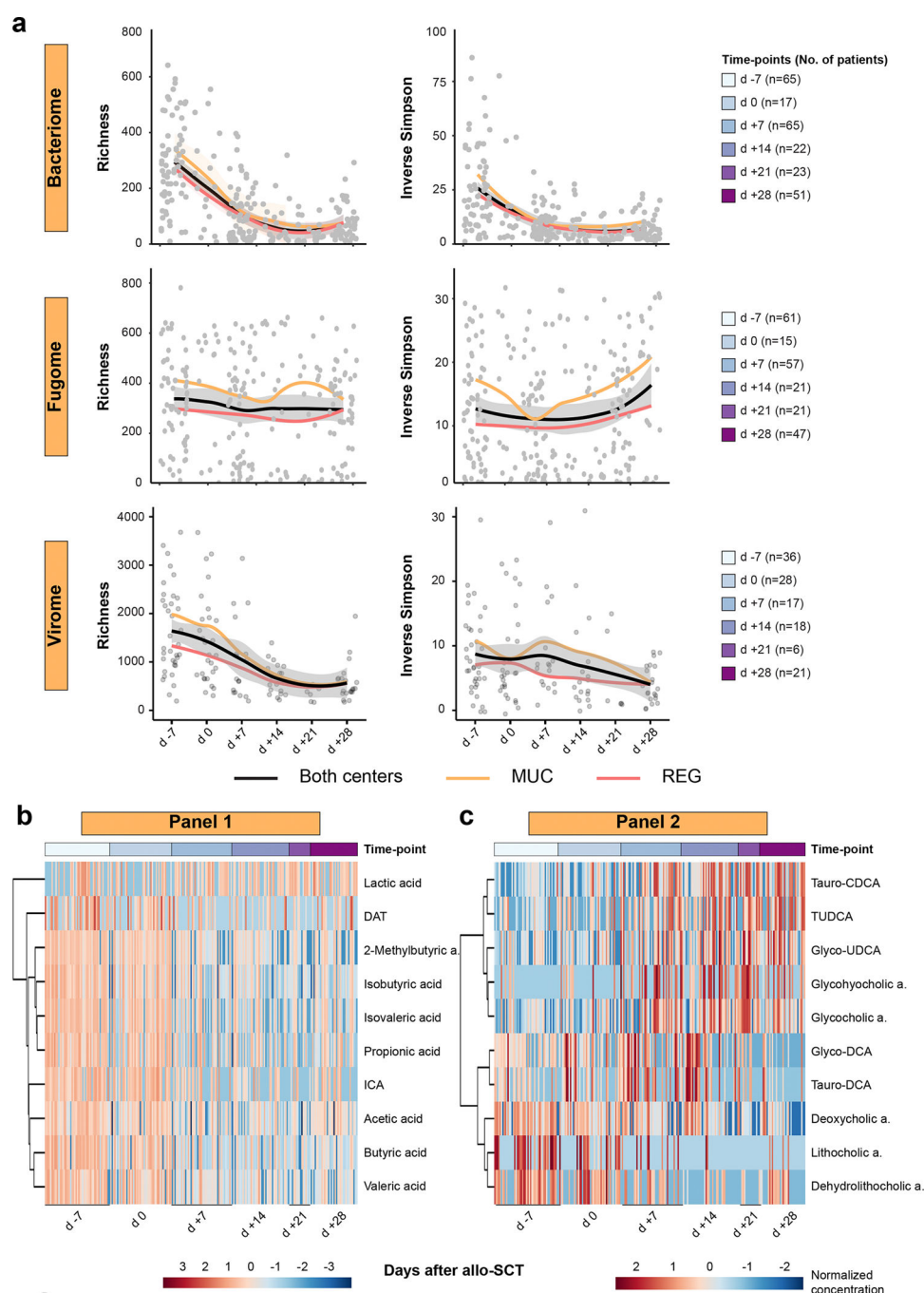


Figure 2:
Longitudinal dynamics of intestinal bacteriome, fungome, virome and metabolome in patients receiving allo-SCT

a) Intestinal microbiome composition in bacteriome (16S amplicon sequencing, n=266 samples), fungome (ITS1 amplicon sequencing, n=245 samples) and virome (viral metagenomic shotgun sequencing, n=138 samples) at time-points relative to allo-SCT (Day 0) by alpha diversity indices (Richness by number of observed zOTUs and Inverse Simpson's diversity index, respectively). Solid lines represent the smoothed conditional

means for the entire cohort (black), Munich (MUC, orange) or Regensburg (REG, red) calculated by locally weighted regression using the locally estimated scatterplot smoothing (LOESS) method. The gray shading indicates the 95% confidence interval of the respective alpha diversity index for the entire cohort. Each individual patient stool sample is plotted as a gray dot superimposed on the graph. The number of patients at each time-point is indicated in the legend.

b) Heatmap of intestinal concentration of microbiota-derived metabolites (in $\mu\text{mol/g}$ dry feces) normalized by log transformation (base 10) in stool samples ($n=269$ samples) of allo-SCT patients ordered by time-points relative to allo-SCT (Day 0). Panel 1 displays short-chain fatty acids (SCFA), branched-chain fatty acids (BCFA), type I-interferon inducing metabolites (IIMs), i.e., desaminotyrosine (DAT) and indole-3-carboxylaldehyde (ICA) as well as lactic acid. Panel 2 shows secondary bile acids (SBA). In each panel, then top 10 metabolites are shown, selected based on highest significance in ANOVA group comparisons between time-points. Clustering based on metabolite expression patterns using the Ward algorithm. Distance measure is Euclidian.

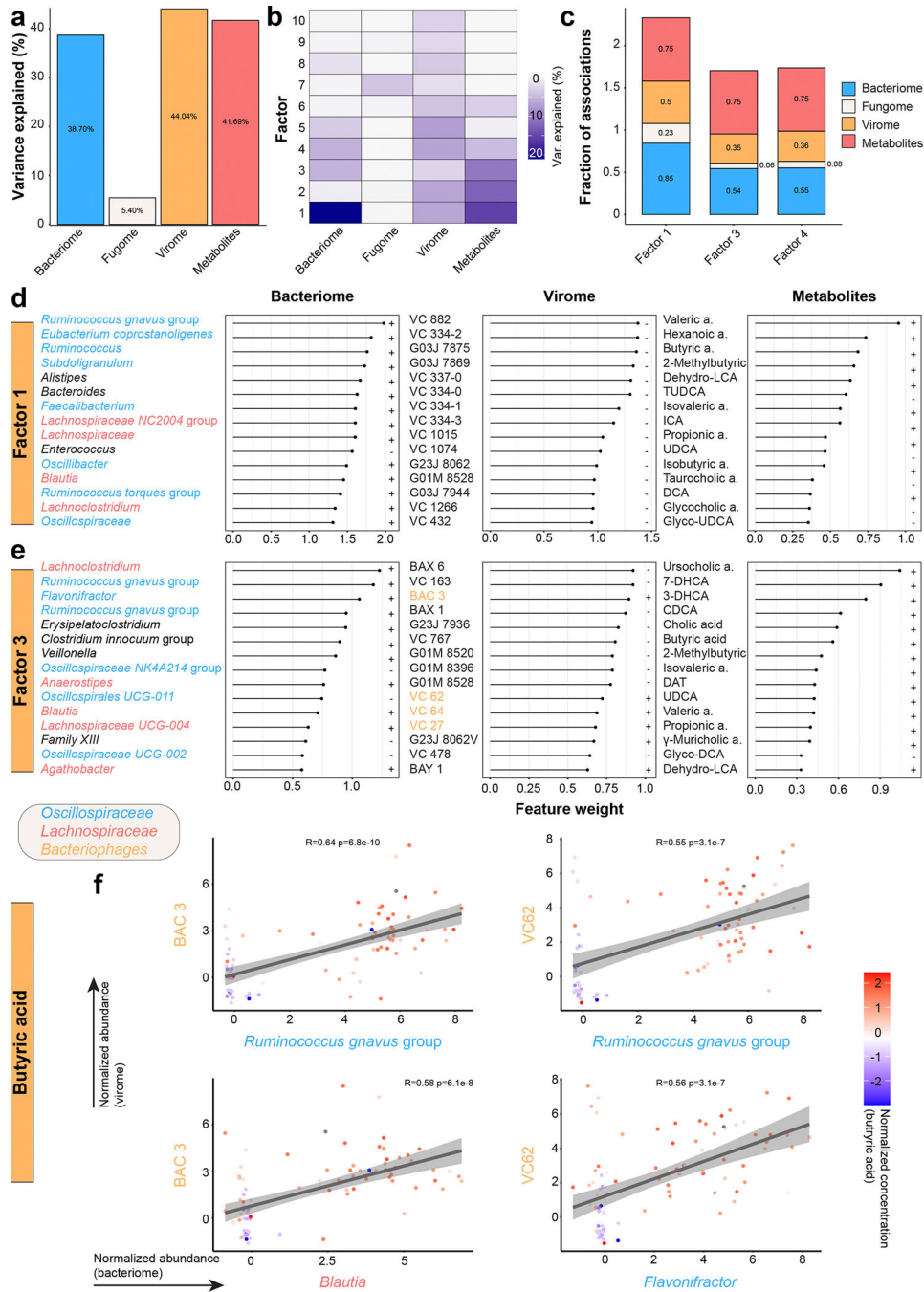
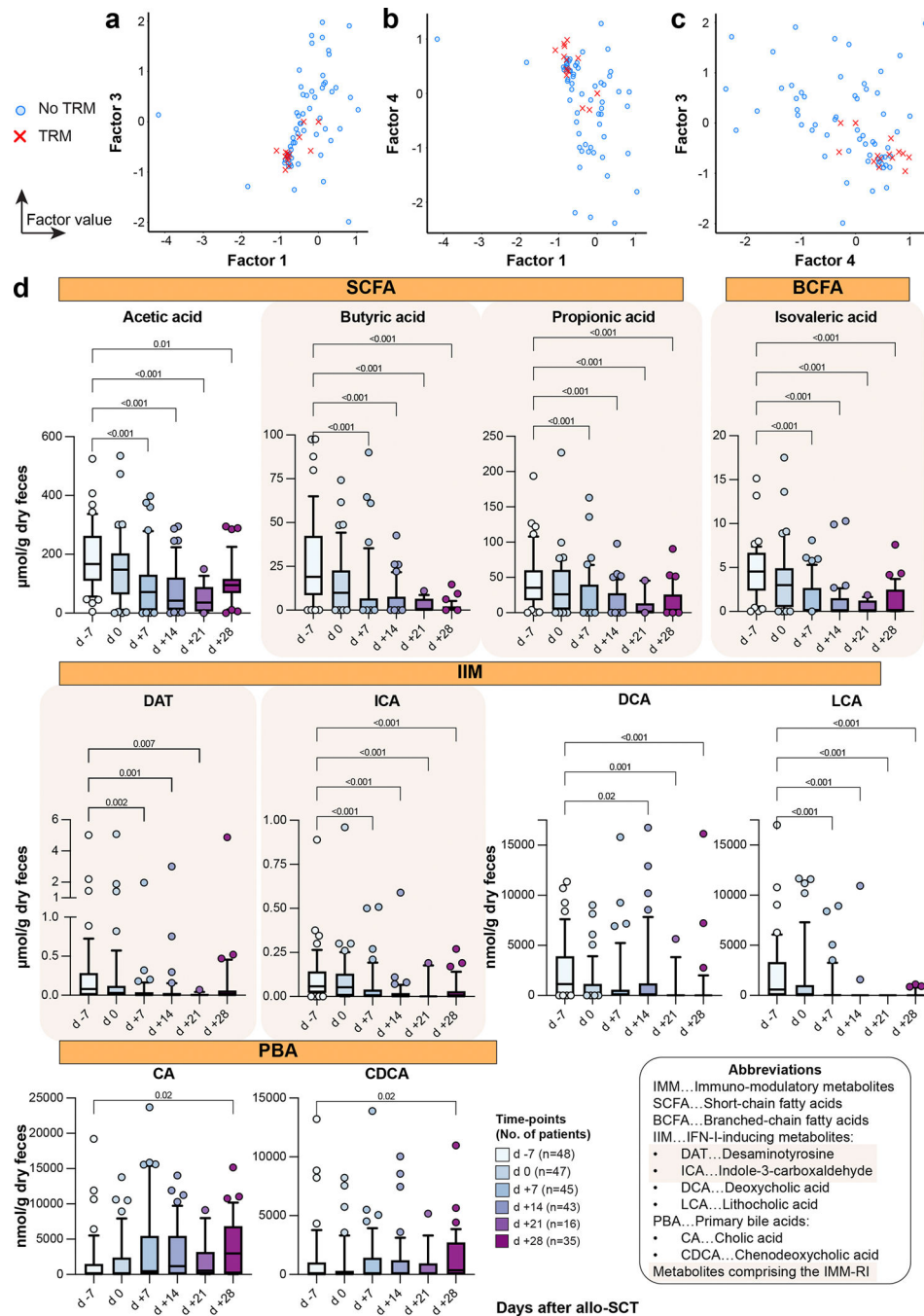


Figure 3: Multi-omics factor analysis (MOFA) identifies bacterial and bacteriophage consortia associated with intestinal immuno-modulatory metabolites (IMMs)

a) Bar plot of the total variance explained by MOFA (in percent) within each omics modality. Exact percentage values are superimposed on the bars.

b) Heatmap of variance explained by individual MOFA-identified Factors, which capture major sources of variability within and across omics datasets. The rows show the percentage of variance explained by a given Factor. The columns indicate the omics modality.

- c) Bar plot showing the fraction of significant associations between the Features of each omics modality and the Factors 1, 3 and 4. Correlations between Features and Factors were analyzed via Pearson's correlation coefficient. P values were corrected for multiple testing via FDR. Statistical significance is called at $p \leq 0.05$. Exact fraction values are superimposed on the bars.
- d) List of high-weight Factor Features within each omics modality in descending order according their corresponding Feature weight. For each Factor, MOFA learns a "weight" as a measure of importance of every Feature contained within that Factor hence enabling the interpretation of the variation captured by that Factor⁴³. A larger weight indicate a higher correlation with that Factor, while the positive or negative sign indicates the directionality of that variation, i.e., "+" indicates a positive association, "-" a negative association. Shown are the top 15 Features in bacteriome, virome and metabolome for Factor 1. Viral taxa are listed by viral cluster name. Blue or pink font coloring indicates membership within the *Oscillospiraceae* and *Lachnospiraceae* families.
- e) As in d) for Factor 3. Orange coloring indicates bacteriophages that were correlated with *Oscillospiraceae* and *Lachnospiraceae*.
- f) Scatter plots of top Features within Factor 3, plotting the normalized abundance of bacteriophages at species level (y-axis) against that of bacterial taxa at genus level (x-axis). Dots represent samples from individual patients at different time points (91 samples from 45 patients). Dots are colored by intestinal levels of butyric acid, or in grey if no butyric acid data was available. Normalized concentrations of metabolites are indicated in the adjacent color legend. Associations between bacterial genera and viral species were analyzed via Pearson correlation and linear regression. The correlation coefficient R as well as the associated p-value is provided in each plot. The regression line is drawn in blue and the 95 % confidence interval of the regression line is shaded in grey.

**Figure 4:**

MOFA Factors are associated with outcome and levels of MOFA-identified IMMs decline progressively after allo-SCT

- a) Factor values of Factor 3 plotted against values of Factor 1 (averaged across Days +7–21). Points are colored and shaped according to the clinical outcome TRM (transplant-related mortality). Blue dots: no TRM at 2-year follow-up (n=58 patients), red crosses: deceased due to TRM at 2-year follow-up (n=15 patients).
- b) As in a), but Factor values of Factor 4 plotted against values of Factor 1.

c) As in a), but Factor values of Factor 3 plotted against values of Factor 4.

d) Levels of intestinal microbiota-derived metabolites at time-points relative to allo-SCT (Day 0) in μmol per gram of dried stool samples measured by targeted mass spectrometry. Metabolites identified as high-weight Features in MOFA Factors 1 and 3 are shown. Shown are (1) SCFA; (2) BCFA; (3) IIM and PBA. The number of patients at each time-point is indicated in the legend. Abbreviations are explained in the text box. Metabolites that comprise the Immuno-modulatory Risk Index (IMM-RI) are highlighted by background shading.

Significance by two-sided Kruskal-Wallis test corrected for multiple testing via Dunn's test of all time-points against baseline (Day -7).

In the box plots, the box ranges from the 25th to 75th percentiles. The line in the middle of the box is plotted at the median. The whiskers are drawn down to the 10th and up to the 90th percentile. Points below and above the whiskers are drawn as individual points. Points indicate individual patient stool samples sampled at the specified time-points.

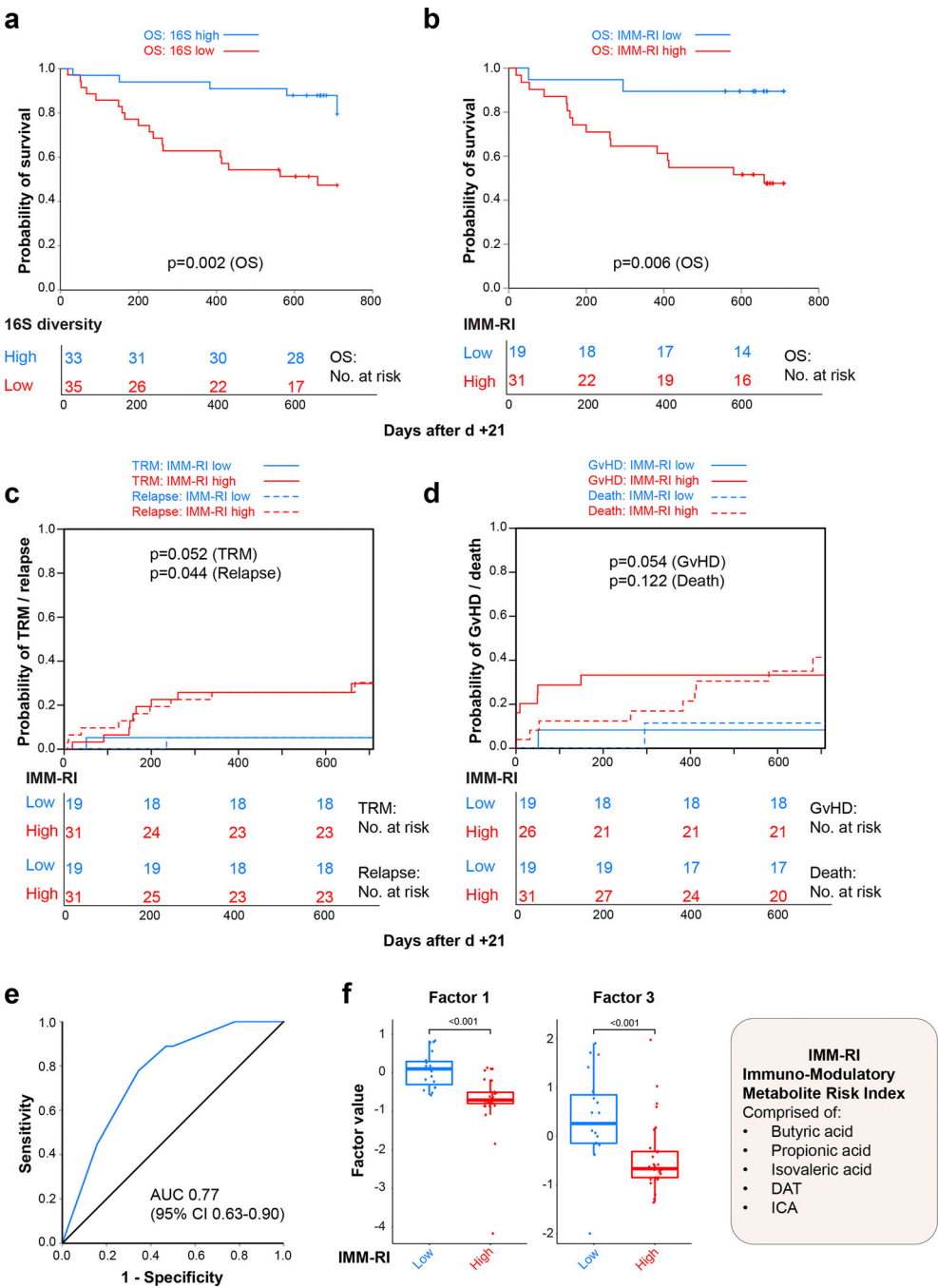


Figure 5:
An Immuno-modulatory Metabolite Risk Index (IMM-RI) is associated with overall survival, relapse, TRM and GvHD
a) 2-year overall survival (OS) after Day +21 stratified according to bacterial alpha diversity. The mean alpha diversity of patient samples at Days +7–21 was calculated and patients were stratified into high and low diversity groups, defined as above or below the center-specific median Inverse Simpson’s diversity index. There were 18 deaths among 35 patients in the low-diversity group (estimated mean survival time 470 (95% confidence interval (CI) 382–

557) days) and 6 deaths among 33 patients in the high-diversity group (estimated mean survival time 658 (95% CI 599–716) days). The number of patients at risk after 0, 200, 400 and 600 days after Day +21 is indicated. Analysis via Kaplan–Meier estimator, significance according to log-rank test. All hazard ratios are reported in Supp. Table 3.

b) OS after a follow-up of 2 years after Day +21 stratified according to high-risk and low-risk IMM-RI, which considers optimal intestinal levels at Days +7–21 of the metabolites indicated in the text box. Optimized thresholds for metabolite concentrations were determined using the Youden Index (Supp. Table 9). Analysis via Kaplan–Meier estimator, significance by log-rank test.

c) Cumulative incidence of relapse and transplant-related mortality (TRM) after 2 years after Day +21 in a competing risks analysis stratified according to high-risk and low-risk IMM-RI. Cumulative incidence functions between groups were tested for equality using the Gray's test, the p-value is indicated.

d) Cumulative incidence of GvHD and its competing risk death after 2 years after Day +21 in a competing risks analysis stratified according to high-risk and low-risk IMM-RI. Statistics as in c).

e) Receiver operating characteristics (ROC) curve indicating sensitivity and specificity for the IMM-RI to predict mortality. The Area under the curve (AUC) is indicated, positive predictive value (PPV) = 0.516 and negative predictive value (NPV) = 0.895.

f) Box plots of Factor values for Factors 1 and 3 (averaged across Days +7–21) in IMM-RI low- vs high-risk patients. The center line corresponds to the median, the box ranges from the 25th to the 75th percentile. Whisker length corresponds to the largest/lowest data point that does not exceed the 75th/25th percentile $\pm$ 1.5-fold interquartile range (IQR). Blue: low IMM-RI (n=19 patients), red: high IMM-RI (n=30 patients). Statistics by two-tailed Mann-Whitney-U-test corrected for multiple testing via FDR.

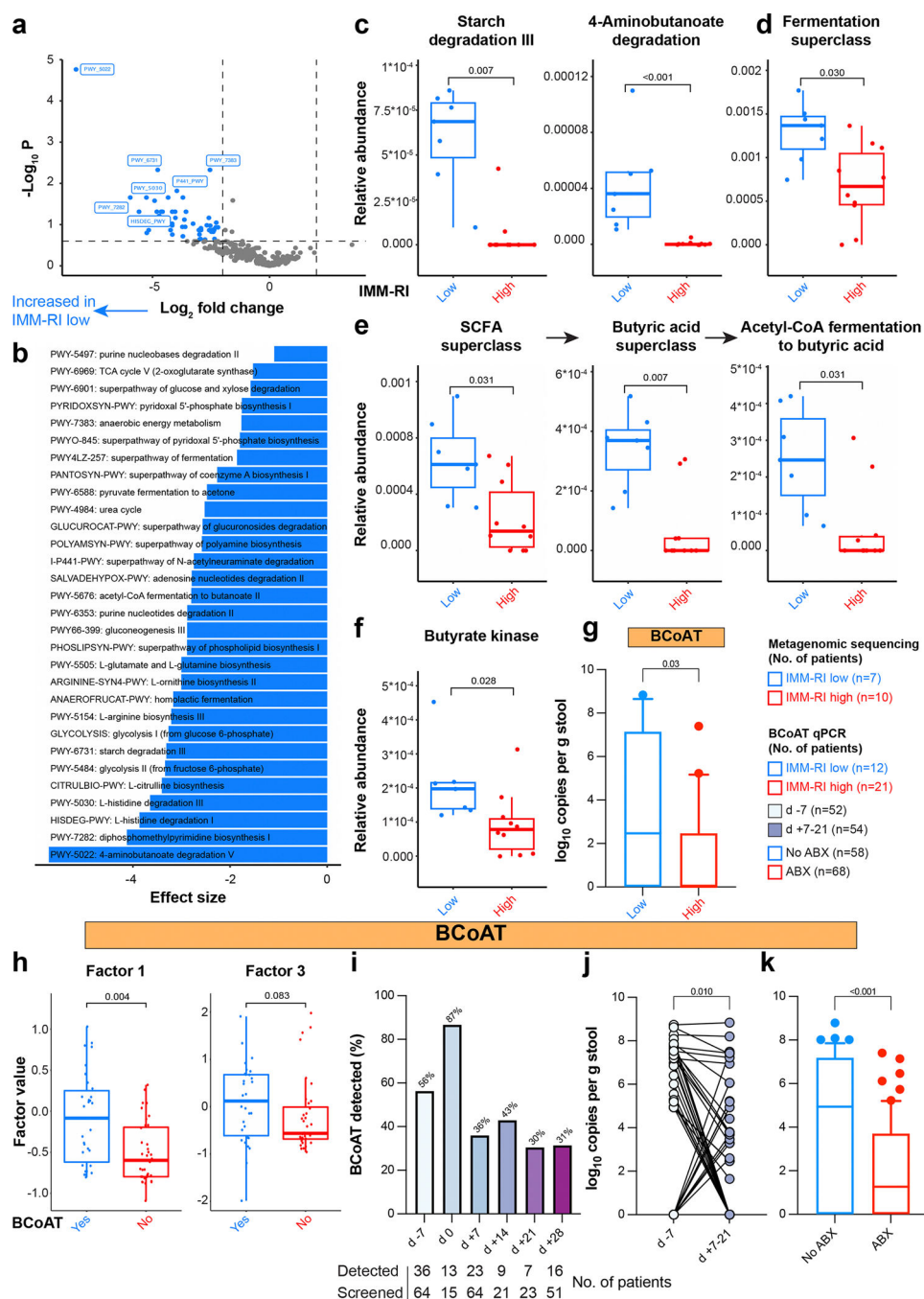


Figure 6:

Microbial SCFA and butyric acid biosynthesis pathways are more abundant in IMM-R low-risk patients

a) Volcano plot of differentially abundant microbial pathways identified by metagenomic sequencing in IMM-RI low- vs high-risk patients. Blue dots indicate pathways which are significantly more abundant in IMM-RI low-risk patients, their BioCyc ID is shown. The x-axis shows fold-change ($\log_2$), the y-axis p-values ($\log_{10}$) adjusted for multiple comparisons via the Benjamini & Hochberg procedure.

- b) Waterfall plot of differentially abundant microbial pathways in IMM-RI low- vs high-risk patients identified with Maaslin2. Only features in the IMM-RI low-risk group were significant after applying abundance and p-value cutoffs ($p \leq 0.01$) and are shown. BioCyc IDs and MetaCyc pathway names are indicated. The x-axis denotes the effect size.
- c) Relative abundance of the specified MetaCyc pathways in IMM-RI low- vs high-risk patients.
- d) Summarized relative abundances within the fermentation superclass in IMM-RI low- vs high-risk patients.
- e) Summarized relative abundances within the SCFA and butyric acid superclasses and the indicated MetaCyc pathway in IMM-RI low- vs high-risk patients.
- f) Relative abundance of butyrate kinase in IMM-RI low- vs high-risk patients.
- g) *BCoAT* copy numbers (in log10) per gram of stool quantified by qPCR according to low- vs high-risk IMM-RI.
- h) Factor values for Factors 1 and 3 (averaged across Days +7–21) according to whether *BCoAT* copies were detected by qPCR (“Yes”) or not detectable (“No”). Blue: detected (n=30 patients), red: not detectable (n=36 patients).
- i) Detection of *BCoAT* copies in patients according to time-point. Bar plots show percentage, exact values are provided. The numbers of patients screened vs those in which *BCoAT* was detected is indicated below.
- j) *BCoAT* copy numbers (in log10) in paired patient samples at Day –7 compared to the peri-engraftment period (samples averaged across Days +7–21).
- k) *BCoAT* copy numbers in paired patients samples before (“No ABX”) and after exposure to ABX (“ABX”).
- For all box plots above: the box ranges from the 25th to 75th percentile. The line in the middle of the box is plotted at the median. The whiskers are drawn down to the minimum and up to the maximum. Samples outside the 1.5-fold IQR were regarded as outliers. In h) the whisker length corresponds to the largest/lowest data point that does not exceed the 75th/25th percentile $\pm$ 1.5-fold IQR. Individual patient samples are plotted as a point superimposed on the graph. The number of patients in each group is indicated in the legend. Statistics: c) to e): 1-sided Wilcoxon test and Benjamini & Hochberg correction for multiple testing; g) 2-sided Mann-Whitney test; h) two-tailed Mann-Whitney-U-test and FDR correction for multiple testing; j) 2-sided Wilcoxon test; k) 2-sided Mann-Whitney test.

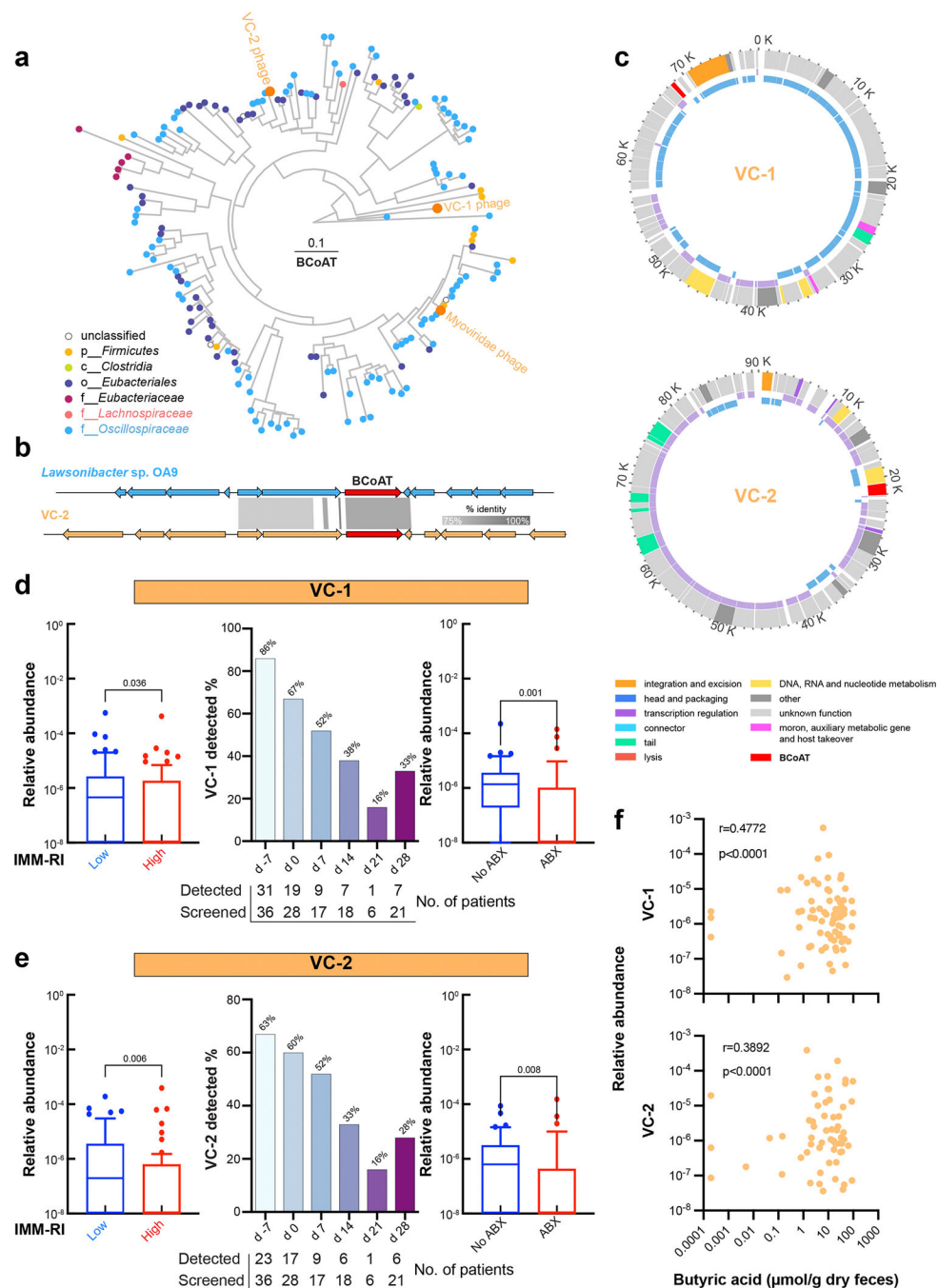


Figure 7:
Detection of *BCoAT* auxiliary metabolic gene encoded in VC-1 and VC-2 bacteriophages
a) Cladogram of bacteriophage-coded *BCoAT* compared against bacterial reference sequences. *BCoAT* was detected in 2 unique viral contigs, VC-1 and VC-2, as well as in a *Myoviridae* sp. bacteriophage. Bacterial taxonomy is indicated in the legend.
b) Gene alignment plot of the VC-2-coded *BCoAT* against a closely related reference sequence in *Lawsonibacter* (an *Oscillospiraceae* family member). The dark-grey shading

indicates identity overlap in percent. A detailed plot is provided in the Extended Data Figure 8.

c) Circle diagram of the VC-1 and VC-2 genomes by viral genome assembly. Genomic regions and their function are indicated in the legend. The *BCoAT* AMG is shown in red.

d) From left to right: Relative abundance of VC-1 according to low- vs high-risk IMM-RI; detection of VC-1 in patients according to time-point; relative abundance of VC-1 in paired patients samples before (“No ABX”) and after exposure to ABX (“ABX”). Bar plots show percentage, exact values are provided. The numbers of patients screened and in which VC-1 was detected is indicated below. In the box plots, the box ranges from the 10th to 90th percentiles. The line in the middle of the box is plotted at the median. The whiskers are drawn down to the minimum and up to the maximum. Samples outside the 1.5-fold IQR were regarded as outliers. High IMM-RI: n=72 samples, low IMM-RI: n=63 samples; “No ABX”: n=40 patients, “ABX”: n=39 patients. Significance by two-tailed Mann Whitney test.

e) As in c) for VC-2

f) Correlation of relative abundance of VC-1 or VC-2 with intestinal butyric acid levels (in $\mu\text{mol/g}$ dry feces) by Spearman’s correlation. Axis is on a log scale. The R- and p-values are indicated in each plot. Number of pairs=130 samples.

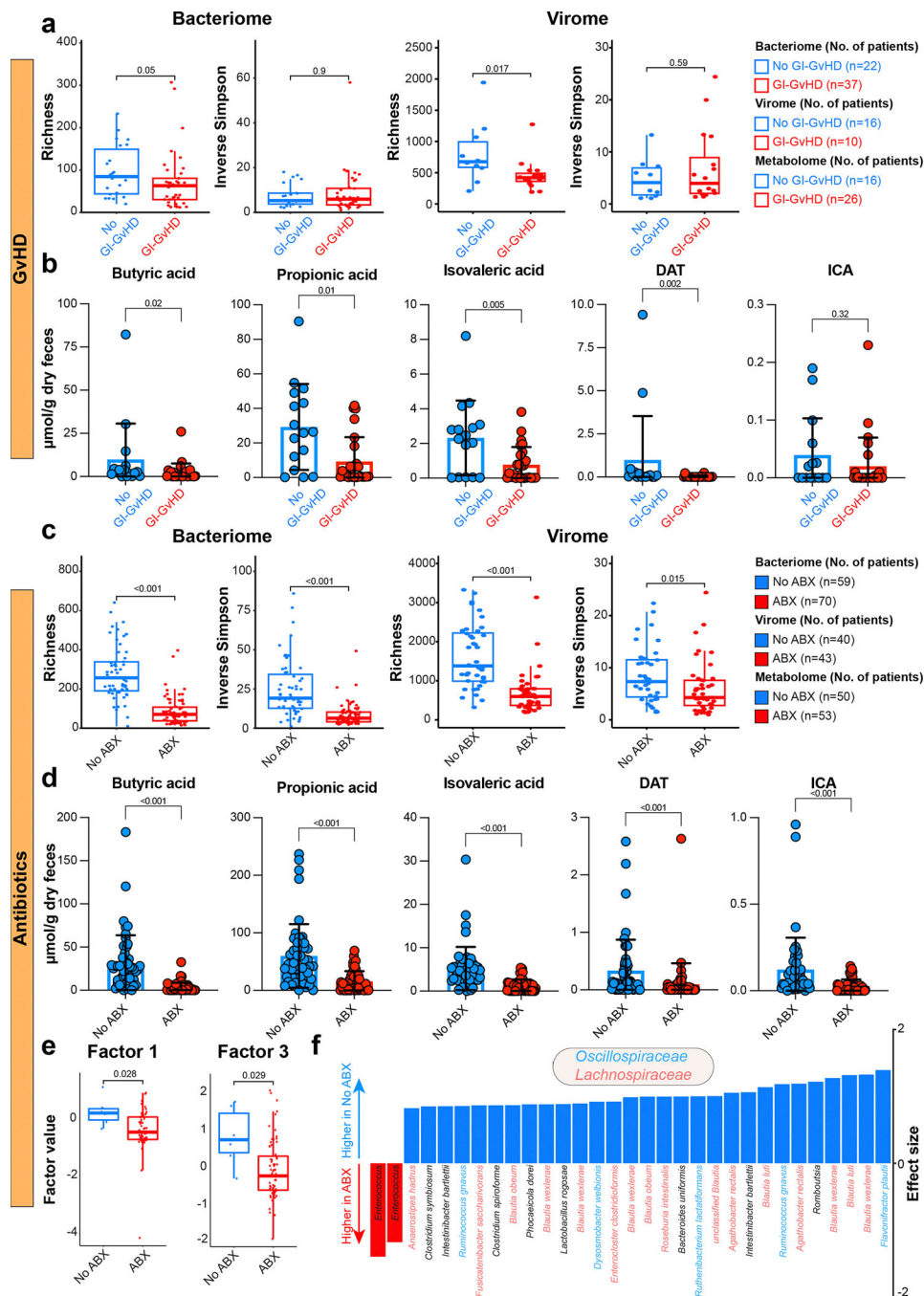


Figure 8:

Onset of GI-GvHD and initiation of antibiotics deplete IMMs

a) Intestinal bacterial and viral alpha diversity (Richness, Inverse Simpson's diversity index) stratified by patients diagnosed with GI-GvHD (GI-GvHD) vs control allo-SCT patients (No GI-GvHD). Significance by two-tailed Wilcoxon rank sum test adjusted for multiple comparisons via the Benjamini & Hochberg procedure. The box ranges from the 25th to 75th percentiles. The line in the middle of the box is plotted at the median. The whiskers are drawn down to the minimum and maximum. Samples outside the 1.5-fold IQR were

regarded as outliers. Each individual patient is plotted as a point superimposed on the graph. The number of patients per group are indicated in the legend.

b) Levels of IMM-RI metabolites (in μmol per gram of dry stool) stratified by patients diagnosed with GI-GvHD vs control allo-SCT patients. Significance by two-tailed Mann-Whitney test. In the scatter-dot plots, the box is plotted at the mean. Error bars indicate standard deviation. Each individual patient is plotted as a point superimposed on the graph. The number of patients per group are indicated in the legend.

c) Intestinal bacterial and viral diversity alpha diversity as in a) in paired patient samples according to antibiotic status: “No Antibiotics (No ABX)” (blue) or “Antibiotics (ABX)” (red). Once a patient was treated with antibiotics, the current and all subsequent samples were classified as “ABX”. Significance by two-tailed Wilcoxon rank sum test adjusted for multiple comparisons via the Benjamini & Hochberg procedure. Box plots as described in a). The number of patients per group are indicated in the legend.

d) Level of IMM-RI metabolite levels in patient samples according to antibiotic status as in e). Significance by two-tailed Wilcoxon matched-pairs signed rank test. Scatted-dot plots as described in b).

e) Box plots of Factor values for Factors 1 and 3 according to ABX (averaged across Days +7–21). Box plot as described in a), except that whisker length corresponds to the largest/lowest data point that does not exceed the 75th/25th percentile $\pm$ 1.5-fold the IQR. Blue: “No ABX” (n=6 patients), red: “ABX” (n=67 patients). Significance by two-tailed Mann-Whitney-U-test corrected for multiple testing via FDR.

f) Waterfall plot of differentially abundant bacterial taxa in patient samples according to antibiotic status. Only significant taxa with a p-value cutoff ($p \leq 0.01$) are shown. Blue bars indicate species that are more abundant in patients not exposed to ABX, red indicates species that are more abundant after ABX exposure. The x-axis denotes the effect size.